

Evaluating Heterogeneity in Factors Influencing the Transport Mode Choice of Madrid's Citizens

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Abstract

This paper investigates the key determinants of transport mode choice among Madrid citizens using Classification and Regression Trees and weighted average SHAP (SHapley Additive exPlanations) values while addressing the heterogeneous nature of urban travel behaviour by applying advanced clustering techniques. The aim is to support urban planners in designing sustainable and inclusive mobility policies. We propose a novel analytical framework comparing suitability of Latent Class Clustering and an enhanced K-Prototypes algorithm which incorporates the Eskin measure to handle mixed-type data. These methods are evaluated using a dataset consisting of socio-demographic, trip-specific, and weather-related variables. Model selection criteria such as AIC, silhouette scores, and cluster stability via bootstrapping are employed to determine the optimal number of clusters. We further assess the quality of clustering through a classification-based approach using LightGBM. Our findings suggest that Latent Class Clustering outperforms K-Prototypes in extracting meaningful consumer segments. Moreover, across all identified clusters, trip distance, vehicle access, and possession of a transport card consistently influence mode choice. The SHAP analysis highlighted the additional impact of weather and rush hours. Thus, the results underscore the importance of tailoring transport policies to different population groups. Our recommendations advocate for targeted discounts, enhanced safety for children and women, and better seat availability during peak hours to promote sustainable and inclusive transport in Madrid.

1 Introduction

The importance of a well-functioning urban transport system cannot be overstated—it enhances city efficiency, supports sustainable consumption, and empowers individuals to make informed travel choices. Consequently, understanding the key factors influencing citizens’ transport mode decisions is vital for policymakers aiming to improve urban mobility and sustainability. Such knowledge becomes especially critical in diverse urban populations, where preferences and behaviours vary widely across social groups.

This research focuses on analysing the transport behaviour of Madrid’s citizens. Its goal is to support urban planners in designing effective strategies and developing targeted incentives to promote public transportation by identifying specific user segments through cluster analysis. Given the highly heterogeneous nature of passenger preferences, a more nuanced and personalised understanding is needed. Madrid, the most populous municipality in Spain and the third-largest metropolitan area in the EU—after London and Paris (World Population Review, 2025)—provides a compelling case study. Over 16% of its population consists of foreign nationals, highlighting a high level of diversity that mirrors the challenges and opportunities many large urban centres face. The research question we will focus on is as follows:

What are the most important factors influencing the choice of transport of Madrid citizens, and how do they differ across consumer segments?

By identifying user segments and their key transport determinants, this research will contribute to the design of sustainable mobility policies, including improvements to bus and metro services and the expansion of cycling infrastructure. It will also offer valuable insights to businesses aiming to align with consumer mobility preferences. Unlike conventional stratification methods—which focus on variables like frequency of use, age, gender, or income—this study applies a more sophisticated segmentation approach that considers both socioeconomic attributes and travel habits simultaneously, representing an innovative contribution to transportation research.

Current research on the determinants of transport choices in Madrid remains limited. Traditional segmentation approaches based on basic demographics fail to capture the complex interplay between socioeconomic characteristics and travel behaviour. Moreover, conventional models such as logit and probit are constrained by rigid assumptions, often overlooking key variables and non-linear relationships. These methods provide only a partial view, leading to imprecise or generalised findings. Additionally, external influences like weather conditions are frequently ignored despite their substantial impact on mobility decisions. For instance, adverse weather may reduce active transport, such as walking or cycling, while favourable conditions can encourage these modes. Integrating climatic data into transport behaviour analysis enhances demand forecasting and enables more adaptive planning (Böcker et al., 2016). To address these gaps, this study proposes using machine learning techniques to segment passengers based on shared perceptions and behaviours. Techniques like Latent Class Clustering (LCC) and Classification and Regression Trees (CART) can uncover complex, non-linear relationships, offering

a more accurate depiction of influencing factors.

This paper presents a comparative analysis of the LCC algorithm and the K-prototypes method to evaluate their effectiveness in clustering mixed-type data. An enhanced K-prototype algorithm implementing the Eskin measure for categorical variables is performed. In order to compare the clustering methods and define the optimal number of clusters, several commonly used methods are used, including the comparison of the average silhouette widths, cluster costs (in terms of distances to the centroid) and the adjusted Rand Index. Furthermore, a classification-based approach is proposed to select the best-performing clustering method, through an analysis of a weighted F1 score and SHAP (SHapley Additive exPlanations) feature importance of a LightGBM model. Additionally, this paper proposes a hybrid modelling approach that integrates Random Forest and XGBoost as an alternative to the traditional CART method. SHAP values enhance model interpretability, enabling a deeper understanding of the most influential factors affecting transport mode choice.

In summary, 5 clusters were identified through LCC (elderly, children, small families, large families, and unemployed citizens). CART and SHAP yielded similar results, highlighting the influence of distance, public transport card possession, and car availability on transport choice decisions. However, a deeper analysis of the features influencing the prediction of the mode of transport was obtained with SHAP. It showed that weather and rush hour travel affect the mode of choice. Moreover, more detailed behavioural patterns across and within groups were derived. For example, the analysis revealed that, contrary to other groups identified, temperature does not seem to significantly affect the choice of mode for elderly citizens. Instead their mobility and accessibility issues are more influential.

Below is a literature review, along with the extensions implemented in this paper. Section 3 describes the data analysed, followed by an extensive description of the methodology in Section 4. Results are then presented in Section 5, and lastly, we draw conclusions from our analysis in Section 6.

2 Literature Review

Tyrinopoulos and Antoniou (2013) conducted a questionnaire-based study to identify key factors influencing transport modal choice in Kalamaria, Thessaloniki, Greece. Using binary probit and structural equation models, they found that parking availability, number of children, crowding, and service unreliability were the most significant determinants of modal choice. A similar study by Madhuwanthi et al. (2016) explored factors influencing transport modal choice in the Colombo Metropolitan Area in Sri Lanka’s Western Province. The study conducted a survey to identify key determinants affecting travel mode selection. The factor analysis results revealed that income, vehicle ownership, safety, and comfort were the most influential factors in modal choice decisions.

The methods above, however, require predefined assumptions about the relationships between independent and dependent variables, which can lead to incorrect service quality likelihood estimations if these assumptions are violated. Nonparametric methods like Artificial Neural Networks (ANN) and CART have been explored to overcome these limitations. ANN has been successfully applied to analyse service quality in metropolitan bus services by Garrido et al. (2014). CART was implemented by de Oña et al. (2016) to assess key service quality factors, given that they provide valuable insights by analysing conditional interactions in data and generating applicable “if-then” rules for policy decisions. Moreover, they offer a standardised measure of the importance of independent variables and enable graphical representation of results, making them a powerful tool for service quality assessment.

Discrete choice models and cluster analysis are alternatives to resolve the issue of heterogeneity among consumer preferences (Hensher et al., 2010). Depaire et al. (2008) and De Oña et al. (2013) have applied LCC analysis to identify traffic accident segments. Unlike clustering methods such as K-means, LCC also provides statistical criteria to determine the optimal number of clusters, enhancing its robustness and interpretability.

Moreover, the heterogeneity across different choices of transport has previously been studied by Haustein and Nielsen (2016) at a European level. Through a k-means clustering algorithm, the authors identified eight clusters representing eight European mobility styles. Next, they clustered the European countries into six segments based on these mobility choices. Their analysis revealed that in Spain, a high fraction of the population either walks or travels by affordable public transport. This paper will investigate whether this is the case in Madrid, Spain’s capital city, by extending de Oña et al. (2016) research where they analysed service quality survey responses of the city of Granada. They did so by first performing LCC and then CART on the overall sample and the identified clusters in order to point out the most important service quality attributes influencing bus passengers’ overall transport evaluations.

This paper’s analysis of the mode of transport choice requires the consideration of both numerical and categorical variables. Traditional clustering techniques such as K-means and K-modes fail to effectively handle both simultaneously (Huang, 1998). Therefore, K-prototype clustering is used, which considers both types of variables and does not require any predefined categories (Ji et al., 2013). As mentioned above, segmenting travellers allows policymakers to improve transportation policies and better allocate budgets towards infrastructure investments. Previously, discrete choice models such as Multinomial Logit and Nested Logit models have been used to analyse transport choices (Ben-Akiva & Lerman, 1985). However, these methods assume specific functional relationships between variables—instead, K-Prototype groups individuals based on similar characteristics. Soria et al. (2020) applied K-Prototype clustering to segment ride-sourcing users in order to analyse the effects of affordability, equity and competition. This suggests the relevance and potential of using K-prototype clustering for transport mode choice analysis to extend the traditional research methods.

Furthermore, we enhance K-Prototypes by using the Eskin measure to calculate the dissimilarity among categorical features, as described by Šulc and Řezanková (2014). This method assigns higher weights to mismatches of features with more categories (Brzezińska & Ogiegło, 2023). In their research, Šulc and Řezanková (2014) concluded that clustering methods implementing the Eskin measure delivered the best within-cluster variability. On the other hand, latent class clustering gave the best results regarding economic interpretation. However, clusters based on the Eskin measure were found to be a good alternative. Hence, the Eskin measure is implemented to K-prototype clustering to analyse whether the performance has improved. We will refer to this method as the Eskin-based K-Prototype (EKP) algorithm.

In addition, we consider Random Forests to examine the influence of features on the choice of transport mode, as they have been found to reduce overfitting and increase predictive accuracy. In fact, Aung et al. (2009) proposed a Random Forest Classifier (RFC) for multi-category web page classification, aiming to organise web content into predefined categories to improve web search quality. Both the Decision Trees classifier and RFCs were applied, and the results showed that RFC achieved over 90% classification accuracy across all categories, outperforming the Decision Tree classifier in every case. Moreover, they also found that combining an RFC with a boosting model achieved an even higher accuracy. Building on this, this study aims to extend the methodology by combining a Random Forest and XGBoost model for a more precise investigation.

3 Data

The primary data source for this study is the Madrid Transport Consortium (CRTM) survey, conducted from February 8, 2018, to June 11, 2018, consisting of 222,744 data points. Participants were primarily contacted through phone calls to answer structured surveys and questionnaires. The CRTM dataset includes trip-specific data (e.g., travel mode, distance, weekday) and socioeconomic data about the participants (e.g., gender, age, education level) as well as household information (e.g., number of cars, number of household members). We have enriched the dataset with meteorological data from the State Meteorology Agency (AEMET), namely the hourly temperature and precipitation. The hourly weather data was then aggregated into trip-specific values based on time-weighted averages, giving higher weights to the hour in which a larger part of the trip duration took place.

The tables of variables used are displayed in the Appendix in Section 8.1, where Table 5 and Table 6 contain the individual and household-specific data used for cluster analysis, respectively. Additional trip-specific variables are included in Table 7, which are also used in the decision trees on top of the individual and household-specific features. Note that the variables with an asterisk have been transformed as described in the Appendix 8.1 to ensure that each variable-level is sufficiently represented and to simplify the analysis allowing for generalization of results. For instance, mode of transport categories such as *private car*, *company car driver*, *rental car driver*, *private car passenger*, *company car passenger*, and *rental car passenger* were consolidated

into a single category labelled “Car”. It is important to note that prior to this grouping, the distribution of responses was examined using a histogram to ensure that frequently occurring categories were not combined inappropriately.

Moreover, dummy variables were created to indicate whether a trip occurred during rush hours, whether the household owns an electric vehicle, and has access to a private parking space. Rush hours are the periods when public transport is the most crowded and roads are most congested. This can influence the choice of transport for individuals who would prefer to avoid such situations (Geng et al., 2023). This research defines the following rush hour periods: 08:00–09:30, 13:30–15:00, and 17:00–19:00. A dummy variable has assigned a value of 1 if at least 75% of a trip’s duration falls within these defined intervals. Rush hours are relevant, as demonstrated by Oakil et al. (2016), who analysed how household and individual-specific characteristics influence the commuting behaviour at rush hours for citizens of the Netherlands. Including rush hour dummy variables allows for an investigation into whether similar patterns hold for residents of Madrid. If confirmed, these insights could support more effective peak demand management, congestion mitigation, and the development of targeted transportation policies (Geng et al., 2023).

As previously mentioned, a dummy variable indicating whether the household owns an electric vehicle is included, given the increasing prominence of restrictions on high-pollution vehicles. Since the start of 2025, Madrid’s Low Emission Zone (LEZ) covers the entire municipal area, which restricts circulation of vehicles without an environmental label (Eurocities, 2025). Exemptions for vehicles registered in Madrid are in place except for the Special Protection LEZs, which only allow access to zero-emission and eco-friendly vehicles.

4 Methodology

Our paper explores how different citizen groups differ in the importance they assign to various factors when deciding on the mode of transport. To achieve that, we first investigate the emerging passenger profiles through Cluster Analysis, deciding on the better segmentation between model-based (Latent Class Clustering) and distance-based approaches (K-Prototypes). Finally, we study the relevance of variables on the choice of the mode of transport, both across the entire sample and within specific groups identified in the cluster analysis. We define the relevance of each variable based on the splitting conditions in the Classification and Regression Trees, as well as on the weighted average SHAP calculated from Random Forest and XGBoost.

While the two-stage approach of segmentation based on LCC and analysis of relevance based on CART splitting criteria has been applied by de Oña et al. (2016), we have extended their methodology in various aspects. We have broadened the investigation by also considering distance-based clustering methods—standard and Eskin-based K-Prototype segmentations—which we have then compared to the LCC through average silhouette width, adjusted rand index, principal-components based three-dimensional plot, and the SHAP values of LightGBM classification

based on the cluster labels. Moreover, we have also elaborated on the decision of the optimal number of clusters using average silhouette width, normalised entropy, and cluster stability based on bootstrapping. Finally, we have extended the CART analysis by considering additional splitting criteria, namely entropy and log-loss, and defined relevance based on SHAPs obtained jointly from Random Forest and XGBoost. The hyperparameter tuning has been achieved through the Bayesian optimisation framework, which is briefly described below.

To maximise the performance of our classifier models (CART, LightGBM, Random Forest and XGBoost), we opt for Optuna. This Bayesian hyperparameter tuning optimisation framework allows for the dynamic construction of a large parameter search space. It prunes unpromising trials for faster results, significantly improving the cost-effectiveness of optimisation (Akiba et al., 2019). The process begins with an objective function which cross-validates a model with trial-specific hyperparameters. Optuna then tries many combinations from the search space. A final model is subsequently trained on the complete data using the best hyperparameters. In CART, the objective function is to maximise the cross-validated accuracy and find the best maximum depth and splitting criterion. Meanwhile, in Random Forest, XGBoost and LightGBM, the objective function is identical, but the aim is to find the best combination of the maximum depth of the tree and the number of trees.

4.1 Latent Class Clustering

Latent Class Clustering is used to split the observations into groups with similar characteristics, attempting to maximise the similarity between in-cluster elements and the dissimilarity between intercluster elements. This particular approach is probabilistic, thus providing probabilities of each observation belonging to each of the clusters. Those values are calculated using Maximum-Likelihood estimates applied to Bayes' rule.

Consider individual i , where $i = 1, 2, \dots, N$. For each individual i , there are P manifest categorical variables $X_{i1}, X_{i2}, \dots, X_{iP}$. Moreover, for each individual i , let us define a latent class variable Z_i that refers to a cluster in a set $\{1, 2, \dots, K\}$. Then, the probability of observing a given $\vec{X}_i = (X_{i1}, X_{i2}, \dots, X_{iP})^T$ is:

$$P(X_{i1} = x_{i1}, X_{i2} = x_{i2}, \dots, X_{iP} = x_{iP}) = \sum_{k=1}^K P(Z_i = k) \cdot \prod_{p=1}^P P(X_{ip} = x_{ip} | Z_i = k) \quad (1)$$

where the equality follows from the law of total probability and an assumption of local independence. Next, we formulate a likelihood function under the assumption of independent and identically distributed $\vec{X}_i$ across the whole sample.

$$L(\theta) = \prod_{i=1}^N \sum_{k=1}^K P(Z_i = k) \cdot \prod_{p=1}^P P(X_{ip} = x_{ip} | Z_i = k) \quad (2)$$

where θ includes all the probabilities to be estimated.

The corresponding likelihood estimates are calculated using the Newton-Raphson or the Expectation-maximization algorithm, denoted below using the *MLE* superscript. Finally, the posterior class membership probabilities for each individual i are calculated using Bayes' rule as follows:

$$\hat{P}(Z_i = k \mid X_{i1} = x_{i1}, \dots, X_{iP} = x_{iP}) = \frac{\hat{P}^{MLE}(Z_i = k) \cdot \hat{P}^{MLE}(X_{i1} = x_{i1}, \dots, X_{iP} = x_{iP} \mid Z_i = k)}{\hat{P}^{MLE}(X_{i1} = x_{i1}, \dots, X_{iP} = x_{iP})} \quad (3)$$

Consequently, an observation is assigned to the cluster for which the above probability is the highest.

4.1.1 Choosing the optimal number of clusters

Given that the Latent Class Clustering relies on Maximum Likelihood estimates, we decided to compute associated information criteria to evaluate the quality of the models. Consequently, we calculated the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Consistent Akaike Information Criterion (CAIC) for a range of cluster sizes, finally plotting them in an attempt to spot the elbow point. Those quality assessment measures balance the model fit and complexity by penalising the log-likelihood function based on the number of parameters which prevents favouring over-fitted models (Akaike, 1998). Those criteria are defined as:

$$AIC = -2\log(\mathcal{L}) + 2\rho \quad (4)$$

$$BIC = -2\log(\mathcal{L}) + \rho \log(n) \quad (5)$$

$$CAIC = -2\log(\mathcal{L}) + \rho[\log(n) + 1] \quad (6)$$

where $\mathcal{L}$ is the model's likelihood, ρ is the number of estimated parameters, and n is the number of observations. These criteria penalise model complexity differently, with BIC and CAIC applying stronger penalties than AIC.

As Anderlucci & Hennig (2014) showed, average silhouette width, a method more traditionally used within the scope of distance-based clustering methods, can also be useful in LCC for determining the optimal number of clusters and deciding whether they have small within-cluster dissimilarities. The average silhouette width measures the cluster validity by assessing the tightness and separation of the clusters (Rousseeuw, 1986). To do so, the similarity of each datapoint to other observations in its own cluster relative to observations in other clusters is quantified, thus arriving at the silhouette score:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (7)$$

where $a(i)$ is the average distance to points in the same cluster and $b(i)$ is the average distance to points in the nearest different cluster. A higher average silhouette is preferred. Given that we are dealing with mixed data in our paper, we need to choose an appropriate distance metric. We opted for Gower's distance; for more details, see D'Orazio (2021). Throughout this paper, we have relied on average silhouette width calculations based on a subsample of 20,000 observations due to computational power limitations.

As a final check for the quality of the clusters, normalized entropy was be calculated using the following formula:

$$E = 1 - \frac{\sum_{i=1}^n \sum_{k=1}^K [-\hat{p}_{ik} \ln(\hat{p}_{ik})]}{n \cdot \ln(K)} \quad (8)$$

where n indexes individual observations, K indexes all the latent classes identified, and $\hat{p}_{ik}$ is the model-estimated probability that the i -th person belongs to the k -th latent class. Values near zero may indicate that the latent classes are not sufficiently well separated, while those closer to one imply clear classification such that observations are confidently assigned to the corresponding cluster.

4.2 K-Prototype Clustering

K-Prototype clustering is a hybrid clustering method for mixed data that extends k-means and k-modes to handle both numerical and categorical attributes. As outlined in Ji et al. (2013), it minimises the cost function:

$$E(U, Q) = \sum_{l=1}^k \sum_{i=1}^n u_{il} d(x_i, Q_l). \quad (9)$$

Here, Q_l is the prototype of the cluster l ; u_{il} ($0 \leq u_{il} \leq 1$) is an element of the partition matrix $U_{n \times k}$; and $d(x_i, Q_l)$ is the dissimilarity measure given by:

$$d(x_i, Q_l) = \sum_{j=1}^m d(x_{ij}, q_{lj}), \quad (10)$$

where

$$d(x_{ij}, q_{lj}) = \begin{cases} (x_{ij} - q_{lj})^2, & \text{if the } l\text{-th attribute is numeric,} \\ \mu_l \delta(x_{ij}, q_{lj}), & \text{if the } l\text{-th attribute is categorical.} \end{cases} \quad (11)$$

The function $\delta(p, q)$ is defined as:

$$\delta(p, q) = \begin{cases} 0, & \text{if } p = q, \\ 1, & \text{if } p \neq q. \end{cases} \quad (12)$$

Here, μ_l is a weight for categorical attributes in cluster l . When x_{ij} is a value of a numeric attribute, q_{lj} is the mean of the j -th numeric attribute in cluster l ; when x_{ij} is a value of a categorical attribute, q_{lj} is the mode of the j -th categorical attribute in cluster l . In order to obtain an appropriate balance, we define μ_l for all clusters as:

$$\mu_l = \mu = \frac{\text{average standard deviation of numerical columns}}{\text{average entropy of categorical columns}} \quad (13)$$

The k-prototypes algorithm is described in detail by Ji et al. (2013) and is outlined in the Appendix in Section 8.2.

4.2.1 Algorithm based on the Eskin measure

To further extend the K-prototype’s methodology, we defined the categorical dissimilarity function using the Eskin measure as outlined in Šulc & Řezanková (2014). Hence, we redefine Equation 11, such that the dissimilarity between two data points regarding their categorical variables is computed as:

$$D(\mathbf{x}_i, \mathbf{x}_j) = \frac{1}{S(\mathbf{x}_i, \mathbf{x}_j)} - 1 \quad (14)$$

where $S(\mathbf{x}_i, \mathbf{x}_j)$ is the similarity between the data points measured as the average similarity of all categories:

$$S(\mathbf{x}_i, \mathbf{x}_j) = \frac{1}{m} \sum_{c=1}^m S_c(x_{ic}, x_{jc}) \quad (15)$$

where m is the number of categorical variables considered.

The Eskin measure specifies that the similarity concerning the c -th variable is computed as:

$$S_c(x_{ic}, x_{jc}) = \begin{cases} 1 & \text{if } x_{ic} = x_{jc} \\ \frac{n_c^2}{n_c^2 + 2} & \text{otherwise} \end{cases} \quad (16)$$

where n_c^2 corresponds to the cardinality of the c -th variable which must be computed beforehand and corresponds to the number of possible values.

4.2.2 Choosing the optimal number of clusters

For standard and Eskin-based K-Prototype clustering, an elbow plot of the costs is plotted, where costs represent the sum of distances of all points to their cluster centroids. The elbow plot often allows the evaluation of the optimal number of clusters, where including an additional cluster leads to an insignificant change in the cost (Lezhnina & Kismihók, 2022).

However, given the often ambiguous interpretation of elbow plots, we complemented this approach by assessing clustering stability. Specifically, for the method with the lower overall costs—either standard K-Prototypes or its Eskin-based variant—we applied a bootstrap resampling technique, following the methodology proposed by Fang and Wang (2012). Assessment of the clustering stability was chosen following the idea that if a clustering solution is of high quality, clustering on random samples should provide similar segments (Wang, 2010). For each k corresponding to the number of clusters, 50 bootstrap samples of size 50,000 were separated from the dataset, and the clustering algorithm was performed on each sample. The adjusted Rand index (ARI) was used to assess the similarity between the original and bootstrapped clusters, which evaluated the cluster stability (Hubert & Arabie, 1985). The ARI adjusts for chance agreement and ranges between -1 and 1, where higher values indicate more stable and consistent clusters (Albatineh et al., 2006).

4.3 Determining the More Suitable Clustering Approach

After performing both LCC and K-Prototypes and deciding on the optimal number of clusters for each, we determined the more suitable clustering approach to carry forward for subsequent analysis. To support this selection, we visualised both segmenting approaches, compared them with each other, and individually evaluated their quality using the identified cluster labels as a dependent variable for the supervised classification. Following the final choice of the more suitable clustering method, CART and ensemble classifiers (Random Forest and XGBoost) were trained across the entire sample and within specific groups identified in the cluster analysis. Based on those results, we arrive at the most important factors influencing the choice of transport mode.

4.3.1 Visualizing and Comparing Cluster Solutions

In order to visualise the defined clusters and their separation, together with the observations belonging to each one, we relied on a common machine-learning approach using Principal Components (Metsalu & Vilo, 2015). Based on the variables used for clustering, the three Principal Components explaining the largest variability in the dataset were used as axes for the 3-dimensional plot. On that plot, all data points were marked with a unique colour corresponding to each cluster. This method allowed us to visualise and understand the cluster structure and spot any emerging patterns or overlaps.

Furthermore, we sought to compare the cluster partitions between the LCC and the K-Prototype Clustering. We relied on the (adjusted) Rand index, which measures the similarity between two clustering methods, with larger values indicating greater similarity between the two approaches. The Rand Index can be calculated using the following:

$$R = \frac{a + b}{\binom{n}{2}} \quad (17)$$

where a is the number of pairs of elements that are in the same subset in the first clustering method and in the same subset in the second clustering method, b is the number of pairs of elements that are in different subsets in the first clustering method and in different subsets in the second clustering method, and n is the number of observations. Moreover, the adjusted Rand Index is a corrected-for-chance version of the Rand Index. To read more about those methods, see Hubert & Arabie (1985).

We constructed a contingency table to compare the resulting cluster assignments between LCC and K-Prototypes systematically. The values in that table represent the number of observations that were assigned to cluster i based on the first method and to cluster j based on the second, with i and j upper bounded by the total corresponding number of clusters. This allowed us to examine the degree of overlap and divergence between the two segmentation methods, providing insights into how consistently individuals are classified across different clustering approaches.

4.3.2 Evaluating Clustering Quality via Supervised Classification

To evaluate the quality and informativeness of the clusters produced by LCC and K-Prototypes, we first compared the average silhouette widths, see Equation 7, with larger values indicating better-defined groupings. Moreover, we adopted a classification-based approach inspired by the framework proposed by Alvarez-Garcia et al. (2024), who introduced a comprehensive method for explainable cluster analysis. In their approach, cluster labels obtained from the clustering method of interest are used as the dependent variable, and an XGBoost classifier is trained to identify the most influential features for distinguishing between clusters, thereby aiding in the interpretation of the key characteristics of each group. This provides insights into the underlying structure and characteristics of the segmentation and showcases which variables are implicitly prioritised by the clustering algorithms in forming the clusters.

Our study applies a similar approach to compare the EKP algorithm with the LCC method. It involves treating the cluster labels as target variables and training a supervised classifier, LightGBM, using the original feature set. If the clustering structure is of high quality, the model can predict the cluster labels with high accuracy. Hence, the distinctiveness is measured by the cross-validated weighted F1 score, where higher values suggest that the clusters are well-separated in the feature space. The F1 score is equivalent to the harmonic mean of precision and recall, providing a balanced measure of classification performance, especially in the presence of class imbalance. Furthermore, the SHAP feature importance from the trained classifier is used to ensure the informativeness of clusters. A complete methodology of SHAP can be found in Section 4.5. It provides a global interpretation of the features on which the classifier is based, revealing the complexity of the clusters. The underlying assumption is that the most important features identified through the classifier’s SHAP values reflect the variables most relevant to the clustering method’s logic and structure.

LightGBM, a gradient-boosting framework that efficiently handles numerical and categorical features, is selected as the classification model, and an 80-20 training-test split is defined. LightGBM provides straightforward integration with SHAP, making it suitable for both performance evaluation and model interpretation. For each clustering method, LCC and EKP, a separate LightGBM classifier is trained to predict the cluster labels. The model is evaluated using stratified cross-validation with the weighted F1 score to assess how well it can distinguish between the clusters. After training, SHAP values are computed to analyse which features the classifier relied on. We will then examine these features to assess whether they are of high quality and whether their influence on the model makes logical sense to finally arrive at a more suitable clustering method.

4.4 Classification and Regression Trees

Classification and Regression Trees are used to build binary decision trees. We aim to explore what variables influence the decision to choose a specific mode of transport both across the whole sample and within the clusters specified in the previous step. Since the target variable is categorical, as it represents the mode of transport, we build classification trees. There are three

steps to building a classification tree. Starting with growing the tree, we recursively partition the target variable based on the splitting criterion, for example, the Gini Index, which measures how pure a node is by summing the probabilities of incorrect classifications for all classes in that node. A node is considered pure when the cases in each subset are all of the same class. The goal is to minimise the Gini Index (i.e., impurity). In other words, the best split is chosen by maximising purity in child nodes, with leaf nodes having lower impurity than the root. The final splits form leaf nodes representing the target variable's values. The process continues until there is no impurity reduction or a predefined minimum number of observations in a leaf is reached. The Gini Index for a given node is defined as follows:

$$G = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk}) \quad (18)$$

We have also considered two more splitting criteria, namely entropy and log-loss:

$$Entropy = - \sum_{k=1}^K \hat{p}_{mk} \cdot \log(\hat{p}_{mk}) \quad (19)$$

$$LogLoss = - \sum_{i=1}^n \sum_{k=1}^K y_{ik} \cdot \log(\hat{p}_{mk}) \quad (20)$$

where K is the number of classes in the target variable, $\hat{p}_{mk}$, the proportion of class k in node m , $1 - \hat{p}_{mk}$, the probability of misclassifying an observation from class k , and $y_{ik} = 1$ if observation i belongs to class k , and 0 otherwise.

The second step consists of pruning the tree according to a cost-complexity algorithm based on removing the branches that add little to the predictive value of the tree to avoid overly large trees and overfitting by trading off model complexity with impurity reduction. For a given α , a subtree T is chosen such that it minimizes:

$$C_{\alpha}(T) = \sum_{m=1}^{|T|} N_m Q_m(T) + \alpha |T|, \quad (21)$$

where $|T|$ is the number of terminal (leaf) nodes in subtree T , $Q_m(T)$, the impurity of node m , N_m , the number of samples in node m , and α the complexity parameter controlling the trade-off between tree complexity and accuracy. Finally, an optimal tree is selected from the pruned trees that minimises the cost-complexity measure across all possible values of α .

4.5 Ensemble Classifiers: Random Forest and XGBoost

In order to investigate how specific features influence the choice of the mode of transport, we consider a hybrid approach of combining two base learners: Random Forest and XGBoost. This allows us to leverage the strength of both tree-based ensemble models. Random Forest is a bagging type of ensemble, where multiple Decision Trees (single classifiers) are built to improve forecast accuracy by combining the results of each individual classifier, completely in-

dependent of each other, during a voting procedure. The conceptual framework of a Random Forest classifier is depicted in the paper by Parmar et al. (2018). XGBoost is a boosting type of ensemble, where Trees are built sequentially, each trying to fix the errors of the previous tree, and predictions are combined based on a weighted sum of predictions. Refer to the paper by Chen and Guestrin (2016) for more details on this model.

Machine learning was long perceived as a black-box model, yet it has become more interpretable due to methods such as SHAP. It enables us to understand the most important features in driving the prediction (Li, 2022). SHAP values are based on game theory, and an importance value is assigned to each feature in a model. They can be used to interpret any machine learning model, including XGBoost and Random Forests. To implement a SHAP model, an explainer object must be created by providing a classification model to compute SHAP values on the testing set.

4.5.1 Weighted Average SHAP Specific to the Mode of Transport

This paper uses a weighted average SHAP from both Random Forest and XGBoost with weights based on model performance, giving a unified view of feature influence across the base learners. For each class of our categorical target variable, the per-sample SHAP values of each base learner are aggregated with the model performance weights. We begin by training both models independently on the entire sample to execute this. The split is as follows: 80% training data and 20% testing data. The macro F1-score, appropriate for multiclass balance, is used to evaluate both models. The weights of their scores are computed to assign a higher SHAP influence to the better model. Performance weights are obtained as follows:

$$w_k = \frac{F1_k}{F1_{\text{rf}} + F1_{\text{xgb}}} \quad (22)$$

where w_k is the weight assigned to model k , where $k \in \{\text{rf}, \text{xgb}\}$, and $F1_k$ is the F1-score of model k .

The next step consists of separately computing per-class SHAP values for each observation for Random Forest and XGBoost on the test sample. Finally, by assembling all our pieces together, we obtain a combined class-specific SHAP for an observation:

$$\phi_{i,j}^{(c)} = w_{\text{rf}} \cdot \phi_{i,j,c}^{(\text{rf})} + w_{\text{xgb}} \cdot \phi_{i,j,c}^{(\text{xgb})} \quad (23)$$

where $\phi_{i,j}^{(c)}$ is the combined SHAP value for observation i and feature j for mode of transport c , w_{rf} and w_{xgb} are the performance-based weights for the Random Forest and XGBoost models respectively, $\phi_{i,j,c}^{(\text{rf})}$ and $\phi_{i,j,c}^{(\text{xgb})}$ are the SHAP values from the Random Forest model and XGBoost model for observation i , feature j , and mode of transport c .

4.5.2 Global Weighted Average SHAP

It is also worth extracting which features are the most influential across all modes of transport, which requires the aggregation of results from Section 4.5.1. This allowed us to create one global SHAP summary plot by calculating the mean absolute SHAP values across all transport choices and observations. This provides the significance metric per feature regardless of the initial SHAP sign. To achieve the global SHAP, the first step consists of summing the absolute values of combined mode-of-transport specific SHAP for an observation obtained in (23) with the addition of weighting by predicted class probabilities of each model. Thus, classes with higher confidence have more influence.

$$\phi_{i,j}^{(2)(c)} = w_{\text{rf}} \cdot p_{i,c}^{\text{rf}} \phi_{i,j,c}^{(\text{rf})} + w_{\text{xgb}} \cdot p_{i,c}^{\text{xgb}} \phi_{i,j,c}^{(\text{xgb})} \quad (24)$$

where $p_{i,c}^{\text{rf}}$ and $p_{i,c}^{\text{xgb}}$ are the predicted probabilities from Random Forest and XGBoost that sample i belongs to class c .

Finally the global weighted average SHAP is:

$$\bar{\phi}_j = \frac{1}{N} \sum_{i=1}^N \left| \phi_{i,j}^{(2)(c)} \right| \quad (25)$$

Important to note is that due to computation power limitations, we decided to use a sub-sample of the testing dataset of 3,000 observations. We have displayed our SHAP results using a summary plot, which accurately identifies important features and quantifies the amount of variable contributions to the model.

5 Results

We begin our findings by determining the number of clusters as implied by LCC and K-Prototypes. For the former, a five-cluster solution is identified based on information criteria and average silhouette width values. For K-Prototypes, relying on the Eskin measure leads to lower costs compared to the standard approach. In addition, based on the segmentation stability analysed using bootstrapping, EKP also found that the optimal number of clusters is five. More details of those findings can be found in Sections 5.1 and 5.2 below. The analysis comparing these two clustering solutions, presented in Section 5.3, suggests that LCC results in a higher quality segmentation and is therefore used in the subsequent analysis. The five identified clusters are described in Section 5.4 and are the following: Elderly, Children, Small Family, Large Family, and Unemployed.

In the second part of our research, Decision Trees (CART) and SHAP values based on Random Forest and XGBoost models are analysed to determine the factors influencing the choice of transport mode. Firstly, the trees resulting from CART were inspected in Section 5.5. The results imply that all clusters mostly rely on distance, car availability, and travel card possession when deciding on their mode of transport. The findings based on the trees' first three or four

levels are somewhat limited, as the depth seems too low to display more significant differences across clusters. Nevertheless, it can be noted that children, the elderly, and unemployed citizens are more likely to walk and use public transport to commute. For those citizen groups, possessing a public transport card is highly influential. On the other hand, families use the car relatively more often, given that they are the group with the largest proportion of individuals with a driving license. While the availability of a car is the key determinant in their choice of transport, owning a public transport card is also highly influential. In fact, for large families, it is the most important factor for trips between 0.609 and 1.592 km, regardless of whether a private car is available or not. In summary, distance, public transport card subscription and availability of a car are the most significant factors affecting the mode of transport chosen.

Given the complexity of interpreting deeper decision trees, the value of our second approach using weighted average SHAP values is particularly highlighted. This analysis, presented in Sections 5.6 and 5.7, supports the key results obtained by CART; thus, agreeing that distance, car availability, and public transport card possession are among the most influential factors for transport choice. However, the results provide further insights into the behaviour and decisions of individuals. They suggest that both weather conditions and travelling during rush hours affect the choice of transport. For all citizen groups, warm weather increases the likelihood of walking. However, this is less prevalent among the elderly, who are more constrained by mobility and accessibility problems. For children, rainy weather increases the likelihood of choosing the metro, whereas for members of small families, commuting to work increases the likelihood of choosing the car. Interestingly, when trips occur during rush hours, small family members prefer to walk, and large family members prefer to travel by metro. In addition, female family members are more inclined to commute by public transport, while males are more likely to travel by car.

5.1 Latent Class Clustering

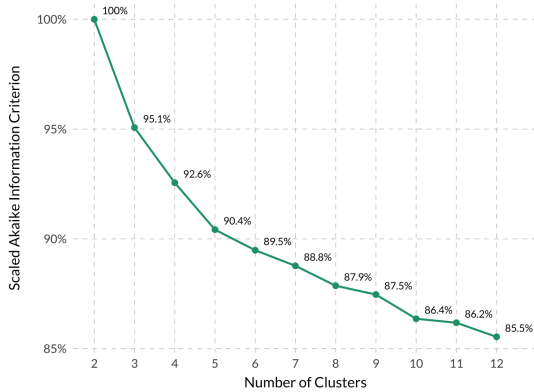


Figure 1: Akaike Information Criterion Across Clusters

Note: Values are scaled by dividing by AIC for two clusters; for k Number of Clusters it is $100\% \cdot \frac{AIC_k}{AIC_2}$

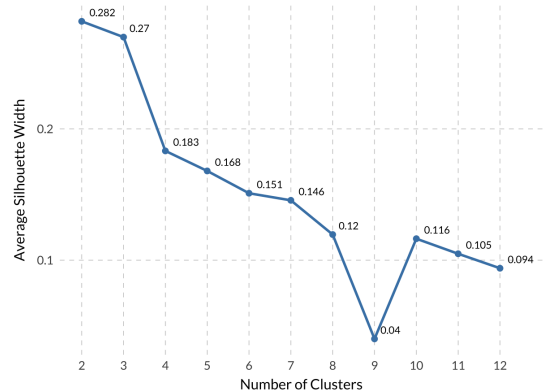


Figure 2: Average Silhouette Width Across Clusters

We first begin with determining the number of clusters suggested by LCC. For each considered number of clusters, 10 optimisation attempts were performed, and the best result was selected

to avoid relying on a local optimum. To decide the number of clusters, we rely jointly on the AIC (BIC and Consistent AIC follow similar patterns) and the average silhouette width. As it can be seen in Figure 1, no clear elbow point indicates the potentially optimal number of clusters. Thus, we also focus on Figure 2 depicting average silhouette width, with larger values indicating, on average, a greater similarity of each data point to its own cluster compared to the nearest neighbouring cluster. The Average Silhouette Width drops significantly at 4 clusters and then continues to decline at a more moderate pace. However, choosing two or three-cluster solutions is nonsensical given that AIC decreases significantly by adding additional clusters. Therefore, **we decide to continue with five-cluster solution**, since it leads to a 2.2 percentage point decrease in AIC compared to the four-clusters as seen on Figure 1, while adding an additional sixth cluster only decreases the information criterion by 0.9 percentage points – which we do not think to be justifiable given the accompanying continued decrease in average silhouette width. Moreover, as seen in Table 1, the corresponding relative change in AIC for the five-cluster solution is more than double that of the six-cluster solution, further suggesting that five clusters should be chosen. The corresponding normalised entropy of five-cluster segmentation is 0.986, suggesting that the clustering exhibits near-perfect separation, with most data points confidently assigned to distinct clusters. See Appendix 8.2 for details on the information criteria and entropy values.

	2	3	4	5	6	7	8	9	10	11	12
Relative ΔAIC	NA	5.19%	2.71%	2.36%	1.05%	0.80%	1.03%	0.46%	1.28%	0.21%	0.75%

Table 1: Relative AIC improvements for cluster solutions from $k = 2$ to $k = 12$.

Note: Each value at k clusters represents the relative reduction in AIC between cluster k and $k-1$, calculated as $100\% \cdot \frac{\text{AIC}_{k-1} - \text{AIC}_k}{\text{AIC}_{k-1}}$. Higher values indicate greater model improvement.

5.2 Standard and Eskin based K-Prototype

Each clustering solution of both standard and Eskin-based K-Prototype algorithms was selected as the best out of 10 runs. This was again performed to avoid relying on local optimum solutions in our analysis, which would be a substantial risk if the clustering algorithm is only run once. A larger number of iterations was not feasible due to the limited available computational power. The costs, as plotted in Figure 3, are scaled with respect to the highest cost, which is the value implied by the standard K-Prototype two-cluster solution. This allows us to see the percentage reduction with respect to this reference scenario. Figure 3 shows that the K-Prototype algorithm that implements the Eskin measure outperforms the standard approach. However, there is no clear elbow point; thus, a stability analysis was performed via bootstrapping on the clusters derived by the EKP algorithm. Figure 4 suggests that the three-cluster solution is the best approach based on this metric, followed closely by five clusters. Given that the five-cluster solution has much smaller scaled costs than the three-cluster segmentation, and since the stability reduction is minimal, **we continue our analysis with five clusters**. This choice aligns with the decision to continue with the 5-cluster solution based on the LCC, thus maximising comparability between the cluster methods in the following analysis (Zaït & Messatfa, 1997).

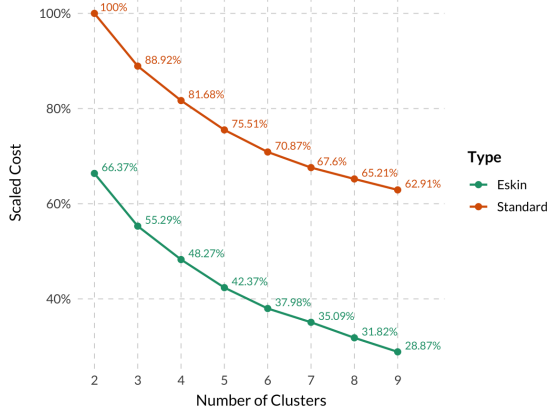


Figure 3: Scaled Cost Elbow Plot

Note: Values are scaled by dividing by the highest cost, i.e. the cost for Standard K-Prototypes with two clusters; for k Number of Clusters and Method i the value is $100\% \cdot \frac{Cost_{i,k}}{Cost_{Standard,2}}$

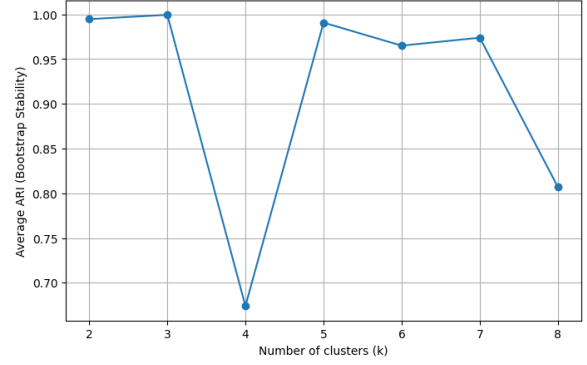
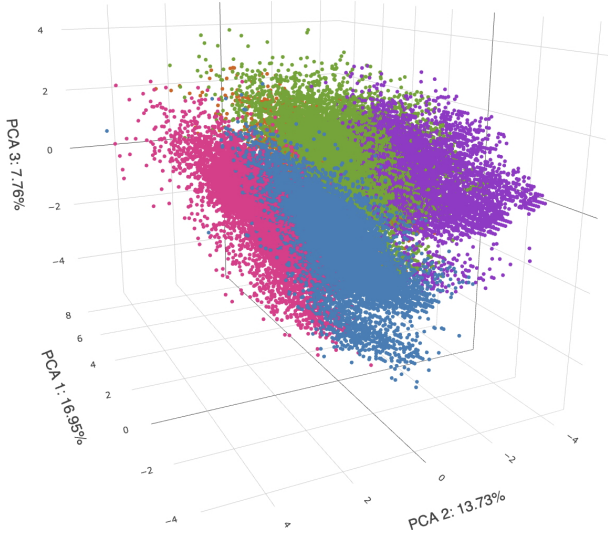
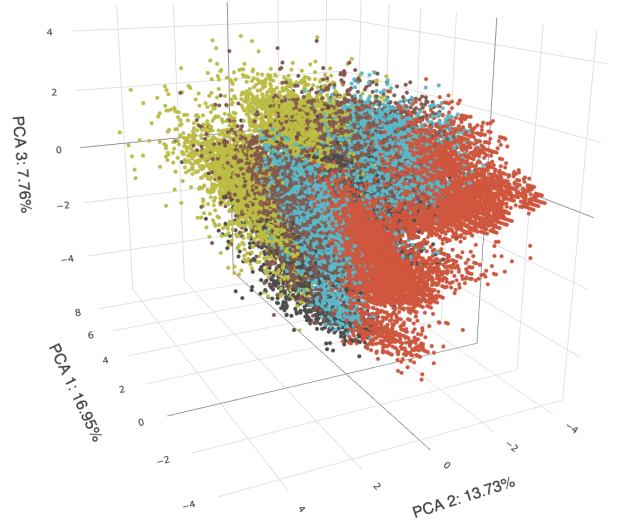


Figure 4: Eskin Cluster Stability

5.3 Selecting the Clustering Solution between LCC and K-Prototypes



(a) Latent Class Clustering for five clusters



(b) Eskin based K-Prototype for five clusters

Figure 5: PCA-based 3D visualizations of the clustering solutions

We first compare the cluster solutions implied by the LCC and EKP. As seen in Figure 5 above, which plots the data points using the three principal components explaining 38.44% variability in total, the two segmentation methods result in very different classes. This finding is further supported by a moderately low adjusted rand index value of 0.2, suggesting that the two clustering approaches produce notably distinct groupings and capture different underlying structures or patterns within the data. Therefore, it is no surprise that the contingency table, see Table 2, suggests that all clusters from LCC, except for cluster 1, are split between all five clusters from the EKP grouping.

		K-Prototypes				
		1	2	3	4	5
LCC	1	0	9,977	0	29,571	106
	2	10,758	5,596	13,543	875	5,266
	3	1,320	26,018	617	20,140	9,745
	4	12,581	1,118	22,344	5	12,200
	5	8,659	9,998	9,378	1,182	11,747

Table 2: Contingency table of LCC and Eskin based K-Prototypes clusters

Note: Value in row i and column j corresponds to the number of observations categorized as part of cluster i for LCC and cluster j for K-Prototypes

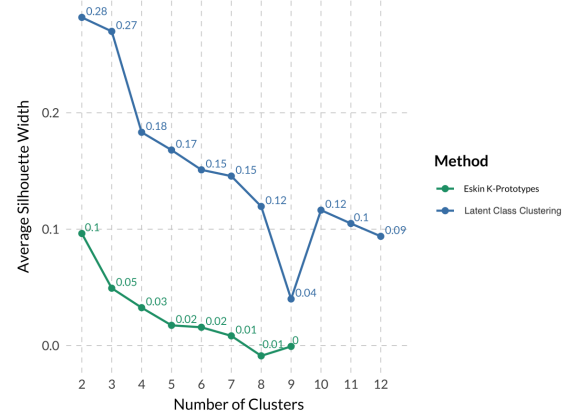


Figure 6: Comparison of Average Silhouette Widths between the methods

To decide on the more suitable approach between the five-cluster LCC and five-cluster EKP, we first consider the average silhouette width; see Figure 6. Assessing the quality of clusters based on a distance metric has been done in previous research by Zait and Messatfa (1997), who relied on average compactness. In this paper, we have decided to proceed with the average silhouette width instead, as it captures both within-cluster tightness and between-cluster separation (Lezhnina & Kismihók, 2022). Furthermore, it is worth noting that comparing two methods implying the same number of clusters using a distance metric is more reliable as it avoids bias that would have otherwise emerged since a larger number of clusters will indicate a closer tightness within a cluster (Zait & Messatfa, 1997). As evident from the results, the values are consistently higher for any cluster size segmentation derived from LCC, which aligns with the clearer separation observed in the 3D principal component plot shown in Figure 5. This suggests that the LCC grouping should be preferred over the one obtained via the EKP. However, to validate our decision further, we compare the approaches using SHAP values from the LightGBM model, treating the cluster labels as the dependent variable of interest.

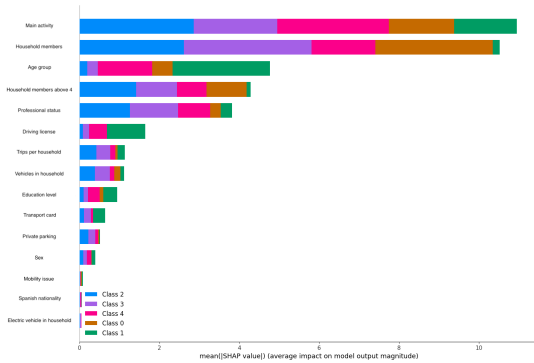


Figure 7: Average SHAP values (in absolute terms) for classifier based on LCC clustering

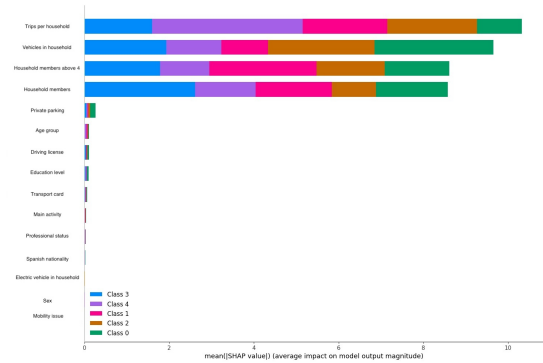


Figure 8: Average SHAP values (in absolute terms) for classifier based on K-prototypes clustering

To classify the LCC clusters, LightGBM relies on 12 features, with at least five of them showing notable importance; see Figure 7. Comparing the results to the average SHAP values of the LightGBM relying on cluster labels from EKP, see Figure 8, variables such as age group, professional status, and main activity play a significant role in cluster classification. Although features such as education level, sex, and possession of a transport card are among the least influential overall, they exhibit relatively greater importance compared to their roles in the EKP clustering. For EKP, the LightGBM classifier primarily relies on just four features: number of trips per household, number of vehicles in the household, number of people in the household, and number of people in the household above the age of four. All other features are of marginal importance, indicated by low average SHAP values. Thus, categorical features played a minimal role in shaping the EKP clusters. Interestingly, this result seems somewhat counterintuitive, as one would expect socioeconomic characteristics such as age group, education level, possession of a driving license, and occupation-related features (main activity and professional status) to be important in distinguishing between different classes of people.

Overall, classifiers for both clustering methods achieve F1 scores close to one, indicating that the clusters produced within each method—EKP and LCC—are highly distinguishable. However, LightGBM requires more features—12 compared to four—to classify the LCC clusters accurately, and several categorical variables gain importance in the process. Notably, socioeconomic characteristics such as age group, professional status, and main activity contribute meaningfully to the segmentation of public transport users, which aligns well with real-world expectations. In contrast, EKP clusters can be almost perfectly classified using just four numerical features, suggesting a simpler and potentially less informative structure. This is not only due to the small number of variables but also because these features might offer limited insights for policymakers interested in understanding the characteristics and motivations of different groups of travellers. This highlights that the clusters produced by LCC are richer and more nuanced, which, together with our findings based on average silhouette widths, provides a strong justification for why **we proceed with the LCC-based five-cluster solution in the subsequent analysis.**

5.4 Final Clusters Identified

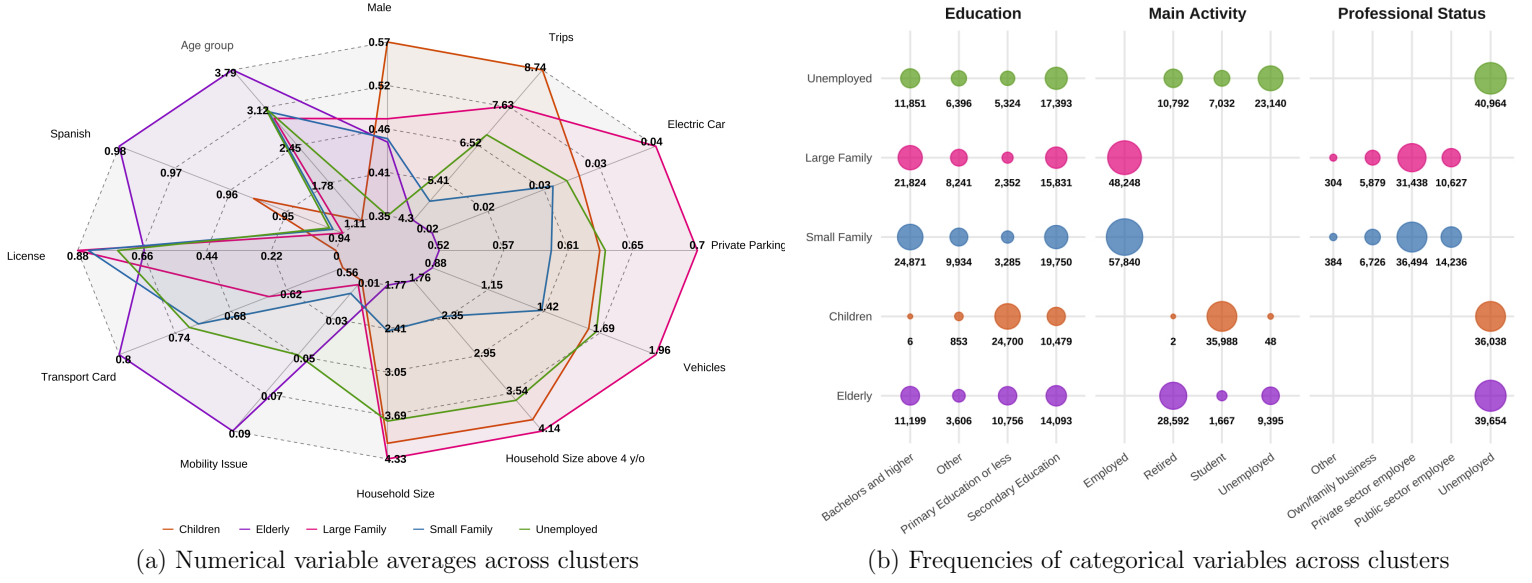


Figure 9: Overview of cluster characteristics

Based on the chosen segmentation method—LCC with five clusters—we have identified the following classes: 1: Elderly, 2: Children, 3: Small Family, 4: Large Family, and 5: Unemployed. As seen in Figure 9a, elderly people are most likely to have a transport card and a mobility issue, while their household size, number of vehicles, and number of trips per week are all the smallest. They are also the least likely to own a private parking or an electric car, suggesting that their reliance on cars in daily transport might be the lowest. On the other hand, children take the largest number of trips per week and live in comparatively larger households. Interestingly, they are the least likely to own a public transport card, potentially suggesting their reliance on public transport could increase if a better children-oriented offer is available. As seen in Figure 9b, they mostly follow primary or secondary education, as expected.

Moreover, those characterised as part of a Large or Small Family have very similar distributions across the categorical variables between each other and differ primarily in the numerical variables. As expected, larger families have, on average, greater household sizes and number of vehicles, and a larger fraction have private parking or an electric car. Nonetheless, both have the highest probability of owning a driver’s license. Lastly, the unemployed are mainly characterised by their professional status and are proportionally mostly female (65%), thus potentially suggesting that citizens of Madrid often follow the old traditional gender roles where the woman manages the household. While they are very likely to own a driver’s license, the number of trips they take per week is among the lowest across the clusters.

5.5 Classification and Regression Trees (CART)

	Full Dataset	Cluster 1: Elderly	Cluster 2: Children	Cluster 3: Small Family	Cluster 4: Large Family	Cluster 5: Unemployed
Splitting Criterion	Log-loss	Log-loss	Log-loss	Entropy	Entropy	Entropy
Maximum Depth	8	7	6	8	6	6
Out-of-sample accuracy	0.6888	0.6937	0.6788	0.6683	0.7051	0.6968

Table 3: Optuna-based best CART models

In what follows, we present the findings from the CART model trained across the whole sample and within the subsamples corresponding to the groups identified using the most suitable clustering method - the five-cluster LCC whose groupings were just described in Section 5.4. Table 3 displays the best hyperparameters chosen using the Optuna Bayesian framework. As can be seen, the splitting criterion used is either Log-loss or Entropy, while the maximum depth of the tree varies between six, seven, and eight. Notably, the decision trees fit the data moderately well, with out-of-sample accuracies ranging from 0.6788 to 0.7051. In the remaining part of this section, we will focus on the key findings on the important variables driving the choice of mode of transport - both across the sample as a whole and within the clusters separately.

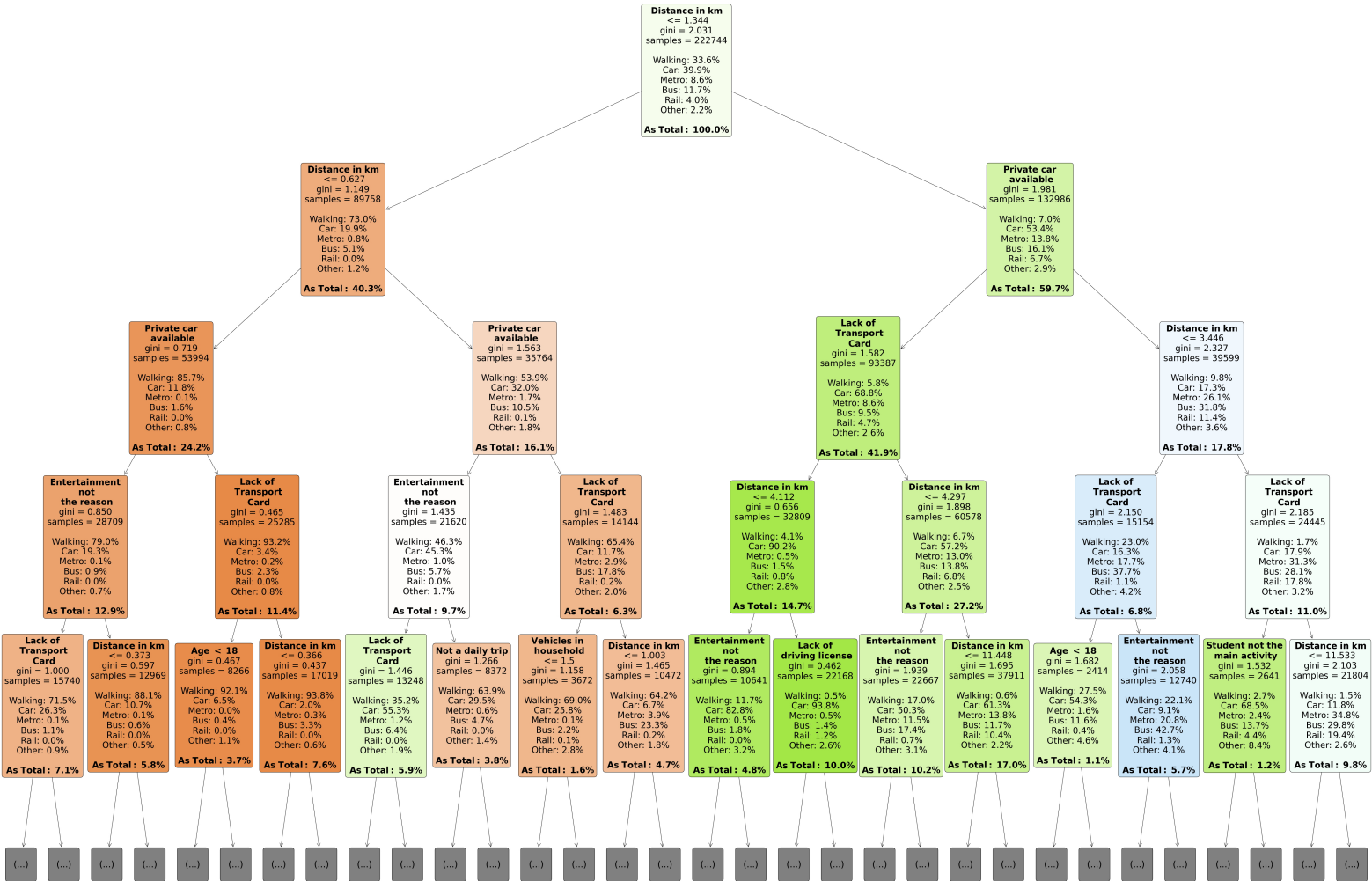


Figure 10: Full Sample Decision Tree Results: Transport Mode Choice

Figure 10 above presents the Classification Tree (CART) model constructed to predict transport mode choice for the full sample. The distance of a trip in kilometres is the most important factor, with those travelling above 1.344 km being most likely to choose the car, while walking is most common for shorter commutes. This finding is intuitive, as longer trips are more likely to require motorised transport. The second most important determinant is the availability of a private car. For journeys longer than 1.344 km, citizens with access to a private vehicle are much more likely to travel by car, whereas those without one tend to rely on public transport. People also base their decisions on the possession of a transport card, with its absence serving as a significant barrier to using metro, rail, and bus services. Moreover, travelling for entertainment greatly increases the chance of walking, which is a leisure-oriented choice. In what follows, the findings of the cluster-specific Decision Trees are outlined to explore the heterogeneity across the citizen groups; see Appendix 8.5 for the plots.

The CART for Cluster 1, representing the elderly, further reaffirms the role of travel distance as a key determinant for transport mode; see Appendix 8.5 Figure 19. The older Madrid citizens are the most likely to walk, while motorised transport is highly probable for trips longer than 1.562 km. Interestingly, for trips between 1.562 and 4.598 km long, travelling by bus or metro is very common, regardless of the availability of a private vehicle. Other key variables influencing decisions include the purpose of the trip (e.g., entertainment), ownership of the transport card, and mobility limitations. Travelling not for entertainment purposes significantly decreases the chance of walking, while having a transport card makes it more likely to go by public transport. Overall, the tree suggests that the elderly predominantly walk for shorter trips and tend to rely on a car for trips above 4.598 kilometres. In other cases, they are inclined to use public transport.

Regarding Cluster 2, which represents children, the primary split is based on the travel distance, indicating the importance of this variable in the choice of mode of transport; see Appendix 8.5 Figure 20. Children mostly rely on walking for shorter journeys, while possessing a public transport card greatly influences the choice between public transport and a car for longer trips. Having such a card considerably increases the likelihood of choosing public transport, while lack of it makes traveling by car the most likely. Concerning shorter distances, children are most likely to walk, and commuting by car becomes more probable for medium distances (between 0.67 and 1.11 kilometres), even more so if they lack a public transport card. What stands out is the fact that the availability of a private car seems to be less important compared to the other clusters. Instead, the number of vehicles in the household is what matters. This likely reflects that children do not drive themselves and depend on others—typically parents or guardians—for car travel. While private car availability indicates whether a vehicle was possibly accessible for a given trip, it does not capture how feasible it actually was for the child to use it. A higher number of vehicles in a household is likely correlated with more people being able to drive it. Lastly, studies being the primary reason of a trip increase the likelihood of traveling by bus, suggesting it is a common choice to commute to school.

Concerning Cluster 3, the small families, walking is more than two times less of a common choice compared to both the elderly and children; see Appendix 8.5 Figure 21. While distance is still the most important factor determining the mode of transport, availability of a private vehicle is a close second—further indicating strong reliance on car transit. Those lacking access to a car lean towards walking in case of shorter trips; while bus, rail and metro are used for longer trips. In contrast, for those with a car available, holding a public transport card still plays a significant role in determining the mode of transport. Individuals without a card are less likely to use public transport, relying instead on walking, car or rail. Additionally, the primary reason for the trip also plays a key role, particularly affecting walking. In fact, people who report entertainment as the purpose of the trip are more likely to walk—which is also a finding for the full sample based on Figure 10. However, when this is not the case, the choice shifts to car, bus or metro. This suggests that small families’ transport decisions are based on constraints such as distance, private car availability, and public transport card ownership as well as the context of the trip.

Large families (Cluster 4) use the car as their mode of transport comparatively most often, i.e. 56.6% of the time; as seen in Appendix 8.5 Figure 22. Analogously to the CART of the small family, the most important variables determining the mode of transport are the distance of the trip and the availability of the private car. The analysis is largely similar, as longer distances lower the probability of walking, the availability of a private car increases the chance of choosing that mode of transport, while the lack of a transport card decreases the probability of going by public transport. However, possessing a transport card seems even more important as it is the most important condition for trips between 0.609-1.592 km, regardless of whether a private car is available. Features like the number of vehicles and household members consistently appear in deeper splits, highlighting their importance in shaping mobility patterns within this cluster.

The unemployed (Cluster 5) are more likely to travel by walking than small and large families, see Appendix 8.5 Figure 23. This appears justified as they are more likely to engage in entertainment activities - whose importance is highlighted in the lower splits of the tree. Again, distance is the primary driver of the choice of transport. Approximately half of trips (48.8%) are shorter than 1.439 km and are dominated by walking, especially when private car access is unavailable. For the remaining 51.2% of longer trips, car availability plays a crucial role in determining mode choice: individuals with access to a car are far more likely to drive, while those without a car are influenced by the availability of a transport card and the trip purpose. The tree highlights that unemployed people rely on public transport for longer trips when the private car is unavailable, especially when the trip is not for entertainment or when they possess a transport card.

5.6 Weighted Average SHAP

Table 4: Model Comparison: Random Forest vs XGBoost across Full Dataset and Clusters

	Full Dataset	Cluster 1: Elderly	Cluster 2: Children	Cluster 3: Small Family	Cluster 4: Large Family	Cluster 5: Unemployed
Accuracy						
Random Forest	0.7690	0.7909	0.7668	0.7909	0.7858	0.8247
XGBoost	0.7882	0.7913	0.7790	0.7913	0.7936	0.8225
F1 Score						
Random Forest	0.5761	0.6080	0.5826	0.6080	0.5727	0.6761
XGBoost	0.6293	0.6135	0.6173	0.6135	0.6115	0.6708
Performance Weight						
Random Forest	0.4780	0.4977	0.4856	0.4977	0.4836	0.5020
XGBoost	0.5220	0.5023	0.5144	0.5023	0.5164	0.4980
Max Depth						
Random Forest	20	20	20	20	20	20
XGBoost	20	20	17	20	20	18
Number of Trees						
Random Forest	450	485	439	485	340	244
XGBoost	246	220	400	220	230	175

Having trained Random Forest and XGBoost with Optuna hyperparameter optimisation independently across the full sample and the clusters from LCC to be able to obtain the weighted average SHAP values, the performance results are displayed in Table 4. In all cases, except for Cluster 5, XGBoost outperformed Random Forest, resulting in higher performance weights. We may now proceed to determine the most influential features for each mode of transport using SHAP summary plots displaying the SHAP values on the X-axis, which quantify how much each feature contributes to the prediction for or against a specific transport mode. The features toward the top have the highest impact. Red indicates a high feature value, while blue represents a low feature value.

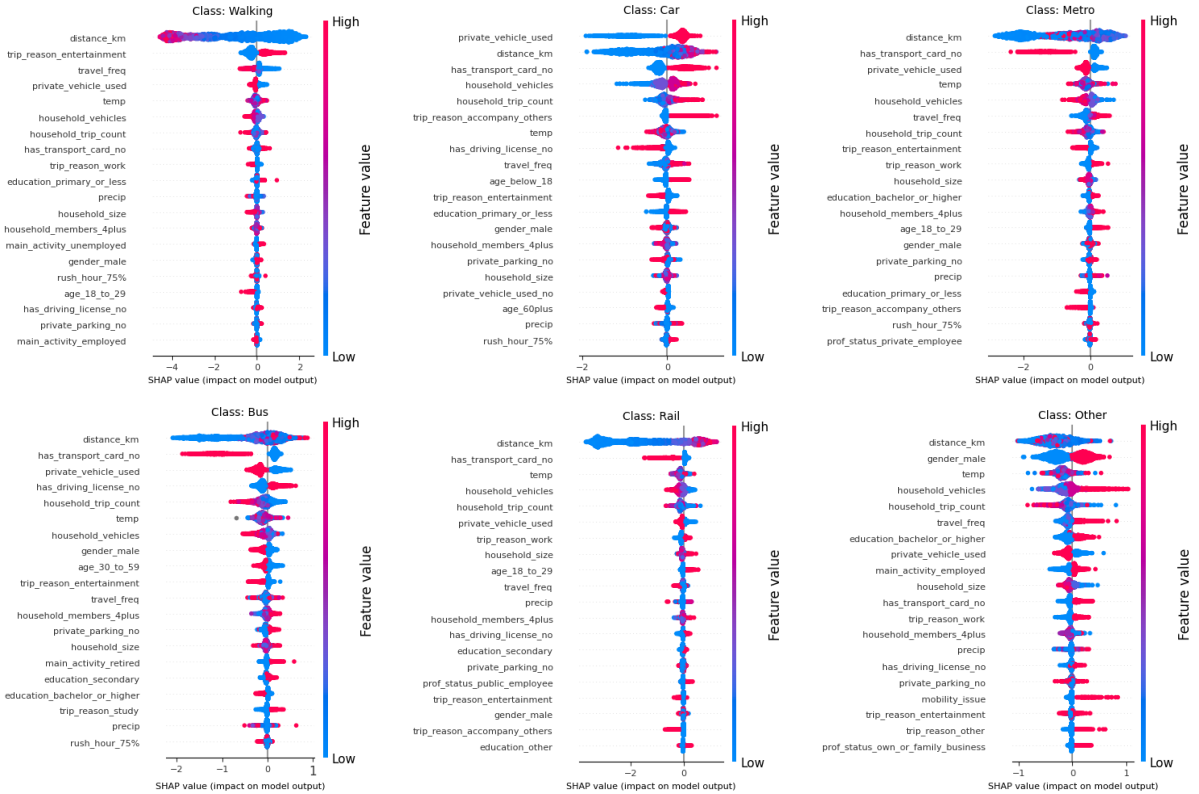


Figure 11: Weighted Average SHAP Plots For Each Mode of Transport On The Full Sample

For the general population in Figure 11, the most influential features for choosing a mode of transport are trip distance, private vehicle availability, public transport card ownership, and travel frequency. Shorter distances strongly increase the likelihood of walking and longer distances push individuals toward cars, rail, or bus. Car use is driven through private vehicle access and lack of a transport card, while metro and bus are preferred when car ownership is limited. Higher temperatures increase walking, while rainfall discourages walking and promotes car and metro usage. The most common trip purpose is entertainment. Individuals with lower education rely more on public transport while vehicle-owning households favour private modes. However, the lack of private parking increases the likelihood of travelling by public transport.

For Cluster 1, representing an elderly group in Appendix 8.6 Figure 24, health and accessibility are the dominant factors influencing the choice of transport mode. For example, walking is largely discouraged when the distance is long, aligning with potential limited mobility among elderly people. Conversely, leisure trips, likely for local outings, more frequent trips, and warmer weather increase their likelihood of walking. Car usage is strongly tied to car availability, longer distances, and the lack of ownership of a transport card. Yet, mobility issues limit the usage of cars. Metro and bus use are preferred for long trips whenever they own a public transport card or lack a car. Rain might deter walking and shift users to the metro and rail. Elderly people tend to use rail for longer and planned trips since distance is the dominant factor. Other modes likely represent alternative solutions for the elderly with reduced mobility, especially when not driving. Temperature does not seem to significantly affect the choice of mode for elderly citizens; hence, their mobility and accessibility issues are more influential.

Cluster 2, identified as children, is mainly influenced by the trip's distance and ownership of a public transport card, see Appendix 8.6 Figure 25. Short distances encourage them to walk, while they opt for car or rail if the travel distances are longer, which is common for school or extracurricular-related travel. Moreover, their choice of walking is more likely in good weather and where car access is low. Warmer temperatures encourage walking, likely due to children being more active when the weather is nice outside. Moreover, children with a public transport card are more likely to commute using the metro, bus or rail. Metro use reflects household decisions for longer trips when in possession of a transport card, especially when children do not have car access. Moreover, younger school children are more likely to use the metro, possibly due to supervised commuting. Rail usage drops without a card. A low feature value for children represented by the variable *age_below_18* suggests that they are less likely to use rail alone. Other modes are used for short and casual trips for fun. It can be deduced that the modes of transport are likely motorcycles or bicycles. Nonetheless, it is limited for younger children, likely due to safety and independence issues. Overall, higher temperatures encourage walking and metro rather than car and other modes of transport, while rain reduces walking and increases car or metro use.

Distance is again the primary determinant for mode choice among the small families represented by Cluster 3 in Appendix 8.6 Figure 26. Walking is preferred for leisure activities, especially

when cars are unavailable, and the weather conditions are favorable. Furthermore, frequent travel increases the likelihood of walking. On the other hand, commuting to work decreases the likelihood of walking. This is likely due to walking being too time-consuming. Small families heavily rely on cars as the main mode of transportation whenever distances are large and they do not have a public transport card nor own multiple vehicles. Not holding a license obviously reduces car usage. Additionally, it can be observed that small families often drive. Metro is used by families without a car who frequently travel, primarily for work when temperatures are higher. Without surprises, holding a public transport card boosts metro, bus and rail use. The bus is used for short-to-medium trips and replaces the car whenever it is unavailable. Working females often use the metro or bus during peak times. However, the positive effect of rush hours on the likelihood of walking is clearer. Precipitation increases the use of metro, car and bus in some cases. Rail is used for long-distance trips, mostly for commuting to work, especially by individuals without a car and with a public transport card. Other transport modes are used for short trips, especially by males when several vehicles are owned by the household, suggesting the ownership of motorcycles and bicycles.

Regarding Cluster 4 representing large families in in Appendix 8.6 Figure 27, distance has again the most significant effect, where they opt for walking, bus or metro for short trips and travel by car or rail otherwise. Moreover, warm weather and frequent leisure outings increase the likelihood of walking, which is likely concerning family walks. When a car is available, large families would rather commute by car, especially in bad weather when coordinating multiple people, or when public transit is less available. Buses are an alternative mode of transportation whenever a car is unavailable, especially for females travelling for work. This suggests that males are more likely to commute to work by car and females by bus. However, travel choices during rush hours discourage the choice of bus. Interestingly, public employees are more likely to replace the car with the metro or bus. Workers, educated families, or adults without cars use rail for longer trips. Other modes are mostly used by younger or male household members for work or by educated individuals without access to a family car. Moreover, rush hour increases the likelihood of using other modes of transport, probably because they opt for a bicycle or motorcycle, which can avoid car traffic.

Finally, unemployed individuals represented by our last cluster in Appendix 8.6 Figure 28 are most likely to walk for short trips, displaying a cost-saving behaviour. The absence of private vehicles increases the dependence on walking, and the ownership of a public transport card increases the likelihood of travelling by bus or metro instead of by car. Car usage among the unemployed is rare, likely due to financial constraints. Furthermore, it only occurs if the household has access to a car and the trip distance is long. Conversely, the metro is a cost-efficient urban solution for unemployed people with a card, particularly for personal errands. Middle-aged users with a moderate education mainly rely on buses in bad weather, presumably because of their affordability and accessibility. Rail is used mainly for longer trips during warm weather or on rainy days by younger or student members in unemployed households who do not have the possibility to drive a car. Lastly, other modes of transport are relied on when no car is

available, mostly for short distances. In addition, warmer weather and mobility issues encourage other transports. This suggests that unemployed people may rely on assisted transport or taxi services. Moreover, the more vehicles the household owns, the more likely they are to opt for other modes of transport, probably because unemployed households tend to have several cheaper modes of transport, such as bicycles or scooters.

5.7 Global SHAP

To conclude the SHAP analysis, a unified feature influence across all modes of transport for each cluster and the full sample is computed from Equation 25, which we named the global SHAP. A bar plot representing the magnitudes of the features is displayed in Figure 12.

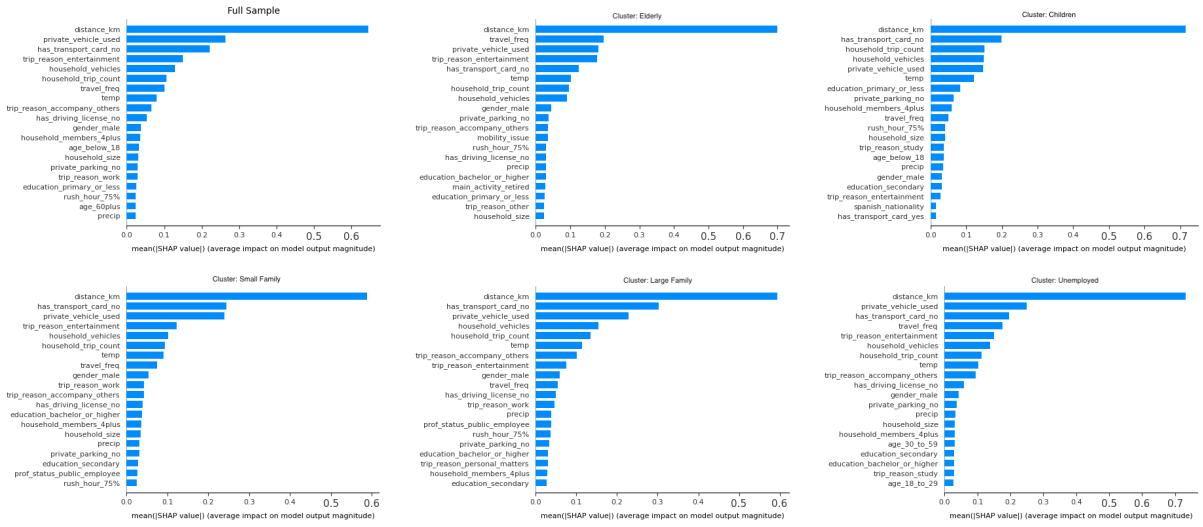


Figure 12: Global SHAP Plots

It can be observed that distance of the trip is always the strongest predictor across all modes. Possession of and access to a private vehicle and a public transport card are consistently high determinants of whether an individual prefers using public transport or a private transport mode. Entertainment, work, and accompanying others are among the primary reasons for the trip, and they recurrently appear in the clusters but vary in rank. Moreover, the weather characteristics of precipitation and temperature have a notable impact on shaping the behaviour of choosing a mode of transport. In addition, commuting during rush hour also significantly affects the choice of transport mode, especially for children (Cluster 2), highlighting the fact that they usually travel to and from school during peak hours. More generally, socio-demographic and trip-specific factors such as gender, age group, and education level vary in importance depending on the cluster, suggesting substantial heterogeneity in behaviour across groups.

6 Conclusion

This paper analysed the most important factors influencing the choice of transport of Madrid’s citizens and how they differ across consumer segments. Firstly, clustering was performed through Latent Class Clustering (LCC), standard K-Prototypes clustering, and enhanced K-prototype clustering implementing the Eskin measure (EKP) to compute the dissimilarity among categorical variables. The choice of clusters was determined by considering several quality measures such as the Aikake Information Criterion, Average Silhouette Width, normalised entropy, and cluster costs (representing the sum of distances of all points to their respective cluster centroids). Moreover, inconclusive elbow plots for the Eskin-based K-Prototype clustering method motivated the use of cluster stability to determine the number of clusters to be chosen. This was performed through bootstrapping and comparing the similarity between a reference clustering solution and the cluster solutions on the bootstrapped samples.

Based on the analysis, the final cluster solutions were LCC and EKP, with five clusters each. These were then visualised and compared through principal component analysis and the adjusted Rand Index (ARI). Those two segmentation methods yielded quite different results, given the ARI of 0.2. Further comparison is undertaken through the classification-based method, LightGBM, where the F1 score and SHapley Additive exPlanations (SHAP) feature importance are analysed. It is evident that the two clustering methods lead to distinct segments and that the quality of the clusters produced through LCC is higher, as indicated by larger average silhouette widths and more relevant variables in the clustering. Hence, we proceeded to analyse the variables that influence the mode of choice for the overall sample and the five clusters identified through LCC - elderly individuals with mobility issues and high ownership of a public transport card; children who travel most likely to school; small and large families mainly differing in number of vehicles and ownership of private parking and electric cars; and unemployed individuals who are mostly female.

The analysis of the Classification and Regression Tree (CART) was run on the full sample and individually on the five clusters identified, highlighting the differences in the variables used to decide on the mode of transport. Overall, the three most important variables across all the citizens are the distance of the trip in kilometres, availability of the private car, and ownership of a transport card. For the elderly, mobility issues and transport card ownership strongly influence reliance on walking and public transport. In contrast, children depend heavily on transport card access, with driving determined mainly by parental availability. Small families exhibit a balanced use of modes, where both card ownership and trip purpose (e.g., entertainment) significantly affect choices. Large families rely more heavily on cars, with household composition (e.g., number of vehicles) playing a bigger role. Finally, the unemployed show the highest use of walking, with entertainment trips and possibly cost constraints shaping their preference for public or active transport modes.

Weighted Average SHAP values based on Random Forest and XGBoost models with Optuna hyperparameter optimisation were evaluated to analyse feature importance. Trip distance,

private vehicle availability, public transport card ownership and travel frequency are the most influential features for the overall sample. However, differences were identified between clusters. The inference about the transport choice and the influential variables is similar to that derived by CART. However, the SHAP graphs provide further insights into other significant variables which CART did not capture. These include the additional variables merged and constructed by us, namely the average trip temperature, precipitation and whether 75% of the trip took place during a rush hour.

Hence, the extensions proposed in this paper have been insightful in determining the factors influencing the citizens of Madrid’s choice of transport and how they differ among segments. This analysis will aid in designing an improved transportation system and advocate for sustainable mobility policies that capture customer heterogeneity. For example, insights about the importance of distance and the possession of public transport cards suggest that the government should carry out campaigns that emphasise the speed of the metro and encourage the subscription to public transport plans. To achieve that, the government could offer discounts for metro and bus subscriptions for frequent commuters such as working adults. Additionally, fares for senior citizens who often rely on private vehicles due to mobility issues and families with children relying on cars to travel to school could also be revised to reduce pollution in Madrid. In order to encourage them to take public transport, assistance for those mobility issues needs to be improved. Other policies include increasing the safety of children commuting alone to school for a safer journey. Furthermore, as a direct result of these policies, the quality of customer service will be enhanced, making aspects such as seat availability a priority concern. This is especially relevant during bus rush hours for working women to ensure a more pleasant experience. In turn, this can also be helpful in the logistics field to optimise the daily schedule to maximise seated customers for a more comfortable ride. Finally, our research can assist Madrid’s Municipal Transport Company, which established the ”Women and Safe Transport” Observatory, in increasing initiatives promoting the safety of women to better assess the situations that may directly affect their safety.

To conclude, several limitations can be pointed out in this paper. Namely, the full sample could not be used for bootstrapping and applying EKP, computing the average silhouette width and the SHAP values. This highlights computational power limitations. Secondly, our analysis did not render any conclusions regarding the influence of electric vehicles; further research should be conducted to study both the impact of electric vehicle possession and travel restrictions recently implemented in Madrid regarding highly pollutant vehicles. Lastly, a possible subsequent research could be to employ double machine learning to formally test for the variables of interest while controlling the nuisance variables.

7 References

- Akaike, H. (1998). Information theory and an extension of the maximum likelihood principle. In *Selected papers of hirotugu akaike* (pp. 199-213). New York, NY: Springer New York.
- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019, July). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (pp. 2623–2631).
- Albatineh, A. N., Niewiadomska-Bugaj, M., & Mihalko, D. (2006). On similarity indices and correction for chance agreement. *Journal of classification*, 23(2), 301-313.
- Alvarez-Garcia, M., Ibar-Alonso, R., & Arenas-Parra, M. (2024). A comprehensive framework for explainable cluster analysis. *Information Sciences*, 663, 120282.
- Anderlucci, L., & Hennig, C. (2014). The Clustering of Categorical Data: A Comparison of a Model-based and a Distance-based Approach. *Communications in Statistics - Theory and Methods*, 43(4), 704–721.
- Aung, W. T., Myanmar, Y., & Hla, K. H. (2009). *Random forest classifier for multi-category classification of web pages*. In *2009 IEEE Asia-Pacific Services Computing Conference (APSCC)* (pp. 372–376). IEEE.
- Ben-Akiva, M. E., & Lerman, S. R. (1985). *Discrete choice analysis: theory and application to travel demand* (Vol. 9). MIT press.
- Böcker, L., Dijkstra, M., & Faber, J. (2016). Weather, transport mode choices and emotional travel experiences. *Transportation Research Part A: Policy and Practice*, 94, 360-373.
- Brzezińska, A. N., & Ogiegło, W. (2023). The similarity of the similarity measures in the context of clustering algorithms for categorical data. *Procedia Computer Science*, 225, 4511-4520.
- Chen, T., & Guestrin, C. (2016, August). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794).
- de Oña, J., de Oña, R., & López, G. (2016). Transit service quality analysis using cluster analysis and decision trees: A step forward to personalized marketing in public transportation. *Transportation*, 43(5), 725–747.
- Depaire, B., Wets, G., & Vanhoof, K. (2008). Traffic accident segmentation by means of latent class clustering. *Accident Analysis & Prevention*, 40(4), 1257-1266.
- D’Orazio, M. (2021). Distances with mixed type variables: Some modified Gower’s coefficients. *arXiv preprint arXiv:2101.02481*.

- Eurocities. (2025, January 24). Madrid’s LEZ clears the air. Eurocities.
<https://eurocities.eu/latest/madrids-lez-clears-the-air/>
- Fang, Y., & Wang, J. (2012). Selection of the number of clusters via the bootstrap method. *Computational Statistics & Data Analysis*, 56(3), 468-477.
- Garrido, C., De Oña, R., & De Oña, J. (2014). Neural networks for analyzing service quality in public transportation. *Expert Systems with Applications*, 41(15), 6830-6838.
- Geng, K., Brands, D. K., Verhoef, E. T., & Wang, Y. (2023). The impact of tradable rush hour permits on peak demand: evidence from an on-campus field experiment. *Transportation Research Part D: Transport and Environment*, 120, 103775.
- Haustein, S., & Nielsen, T. A. S. (2016). European mobility cultures: A survey-based cluster analysis across 28 European countries. *Journal of Transport Geography*, 54, 173-180.
- Hennig, C. (2007). Cluster-wise assessment of cluster stability. *Computational Statistics & Data Analysis*, 52(1), 258-271.
- Hensher, D. A., Mulley, C., & Yahya, N. (2010). Passenger experience with quality-enhanced bus service: The tyne and wear ‘superoute’ services. *Transportation*, 37, 239-256.
- Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data mining and knowledge discovery*, 2(3), 283-304.
- Huang, Z., & Ng, M. K. (1999). A fuzzy k-modes algorithm for clustering categorical data. *IEEE transactions on Fuzzy Systems*, 7(4), 446-452.
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of classification*, 2, 193-218
- Ji, J., Bai, T., Zhou, C., Ma, C., & Wang, Z. (2013). An improved k-prototypes clustering algorithm for mixed numeric and categorical data. *Neurocomputing*, 120, 590-596.
- Lezhnina, O., & Kismihók, G. (2022). Latent class cluster analysis: selecting the number of clusters. *MethodsX*, 9, 101747.
- Li, Z. (2022). Extracting spatial effects from machine learning model using local interpretation method: An example of SHAP and XGBoost. *Computers, Environment and Urban Systems*, 96, 101845.
- Madhuwanthi, R. A. M., Marasinghe, A., RPC, J., Dharmawansa, A. D., & Nomura, S. (2016). Factors influencing to travel behavior on transport mode choice-A case of Colombo metropolitan area in Sri Lanka. *International Journal of Affective Engineering*, 15(2), 63-72.
- Madrid. Madrid Population 2025. (n.d.).
<https://worldpopulationreview.com/cities/spain/madrid#overview>

- Metsalu, T., & Vilo, J. (2015). ClustVis: a web tool for visualizing clustering of multivariate data using Principal Component Analysis and heatmap. *Nucleic acids research*, 43(W1), W566-W570.
- Oakil, A. T. M., Nijland, L., & Dijst, M. (2016). Rush hour commuting in the Netherlands: Gender-specific household activities and personal attitudes towards responsibility sharing. *Travel Behaviour and Society*, 4, 79-87.
- Oña López, J. J. D., López Maldonado, G., Mujalli, R. O., & Calvo Poyo, F. J. (2013). Analysis of traffic accidents on rural highways using Latent Class Clustering and Bayesian Networks. *Accident Analysis & Prevention*, 51, 1-10.
- Parmar, A., Katariya, R., & Patel, V. (2018). A review on random forest: An ensemble classifier. In *International conference on intelligent data communication technologies and internet of things* (pp. 758-763). Springer International Publishing.
- Redman, L., Friman, M., Gärling, T., & Hartig, T. (2013). Quality attributes of public transport that attract car users: A research review. *Transport policy*, 25, 119-127..
- Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics*, 20, 53-65.
- Soria, J., Chen, Y., & Stathopoulos, A. (2020). K-prototypes segmentation analysis on large-scale ridesourcing trip data. *Transportation Research Record*, 2674(9), 383-394
- Šulc, Z., & Řezanková, H. (2014). Evaluation of recent similarity measures for categorical data.
- Tyan, J., Liu, S.-C., & Fu, J. Y. (2024). How environmental, social, and governance implementation and structure impact sustainable development goals. *Corporate Social Responsibility and Environmental Management*, 31.
- Tyrinopoulos, Y., & Antoniou, C. (2013). Factors affecting modal choice in urban mobility. *European Transport Research Review*, 5, 27-39.
- Wang, J. (2010). Consistent selection of the number of clusters via crossvalidation. *Biometrika*, 97(4), 893-904.
- Zait, M., & Messatfa, H. (1997). A comparative study of clustering methods. *Future Generation Computer Systems*, 13(2-3), 149-159.

8 Appendix

8.1 Data: Variables

Individual Specific	
Variable:	
Gender	1. Male / 2. Female
*Age in 2018	1. < 18
	2. 18 – 29
	3. 30 – 59
	4. 60+
Do you have Spanish nationality?	1. Yes/ 2. No
*Do you have a driving license?	1. Yes/ 0.No
*What is your highest level of education?	1. Primary Education or less
	2. Secondary Education
	3. Bachelors and higher
	4. Other
*What is your main activity?	1. Employed
	2. Retired
	3. Unemployed
	4. Student
*What is your professional status?	1. Public sector employee
	2. Private sector employee
	3. Own/family business
	4. Other
	5. Unemployed
*Do you have any of these transport cards?	1. Yes / 0. No
Is your transport subscription currently loaded?	1. Yes / 2. No
Do you have any reduced mobility issues?	1. Yes / 2. No

Table 5: Individual Specific Data For Cluster Analysis

Household Specific	
Variable:	
How many people live in your household, including yourself?	
How many people in the household are 4 years old or older?	
Total number of vehicles in the household	
*Do you have a private parking?	1.Yes / 0.No
*Do you have an electric/hybrid mode of transport?	1.Yes / 0.No
Number of Trips per Household	

Table 6: Household Specific Data For Cluster Analysis

Trip Specific	
Variable:	
*Primary mode of transport - user Y of interest	1. Walking
	2. Car
	3. Metro
	4. Bus
	5. Rail
	6. Other
*Primary Reason for the trip	1. Work
	2. Study
	3. Entertainment
	4. Accompanying another person
	5. Personal matters
	6. Other
Distance, in kilometers, from origin to destination	
*Rush Hour time of the day (Transformation of the time variable)	1. Yes
	0. No
Travel frequency	1. Daily, Monday to Friday
	2. Between 2 and 4 workdays per week
	3. Less than 2 workdays per week
	4. Occasionally
	5. First time making this trip
Do you have a private vehicle for this trip?	1. Yes / 2. No
Average temperature on reference day (merged data)	
Average precipitation on reference day (merged data)	

Table 7: Trip Specific Data For Decision Trees

The variables marked with an asterisk above previously had the possible values below but were transformed to reduce the number of alternatives by taking into account the frequency of the answers. The distribution of the alternatives are shown on the corresponding histograms.

- Age in 2018: before was a continuous variable transformed into categories.

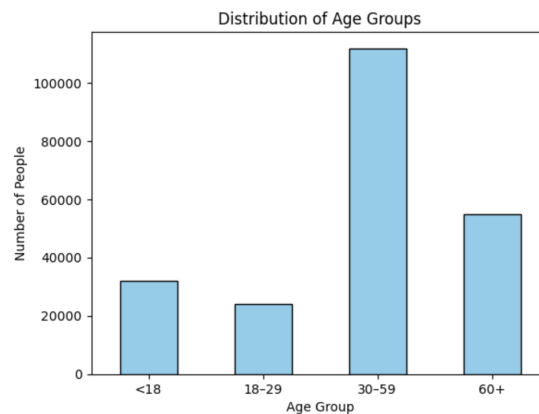


Figure 13: Age groups histogram

- Do you have a driving license?
Before, respondents answered what type of driving license they owned, but we transformed it into a binary variable. The previous answers were:
 - No license (No license)

- Only a moped license (License)
- Motorcycle license A1/A2 (License)
- Car license (B) or higher (C, D, or E) (License)
- Motorcycle and car license (License)

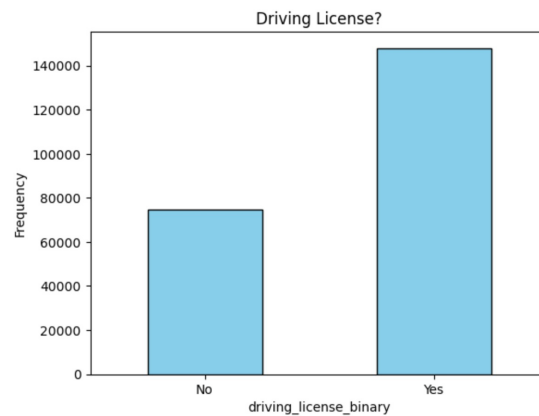


Figure 14: Driving license histogram

- What is your highest level of education?
 - Less than Primary (Primary education or less)
 - Primary Education (Primary education or less)
 - First stage of Secondary Education (Secondary education)
 - Second stage of Secondary Education (Secondary education)
 - Post-secondary non-tertiary education (Other)
 - Higher Vocational Training or equivalent (Other)
 - University degrees, master's, or doctoral studies (Bachelors and higher)

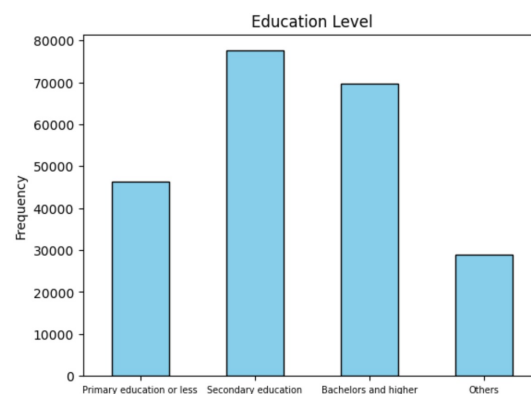


Figure 15: Highest level of education histogram

- What is your main activity?
 - Employed (Employed)

- Employed and studying (Employed)
- Retired/Pensioner (Retired)
- Unemployed, previously worked (Unemployed)
- Unemployed, seeking first job (Unemployed)
- Student (Student)
- Caregiver for family members (Unemployed)
- Unpaid domestic work (Unemployed)
- Other (Unemployed)

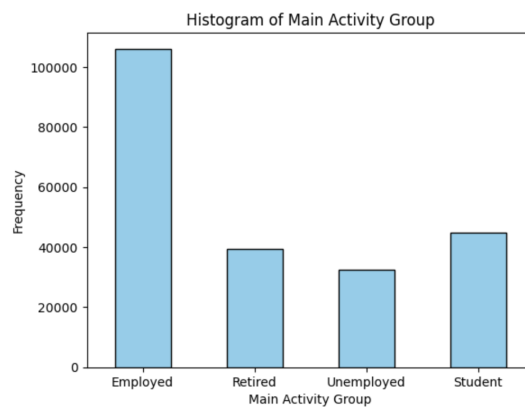


Figure 16: Main activity histogram

- What is your professional status?
 - Public sector employee (Public)
 - Private sector employee (Private)
 - Self-employed or business owner (Business)
 - Helping in a family business (Business)
 - Other employment situation (Other)

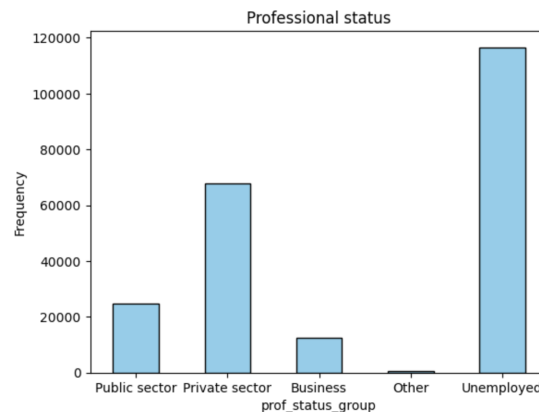


Figure 17: Professional status histogram

- Do you have any of these transport cards?

Since we are only interested in whether an individual has a public transport card or not, we binarized the variable. The possible answers before were:

- Public Transport Card (TTP) (Transport card)
- MULTI Card (Transport card)
- Both cards (Transport card)
- None (No transport card)

- Do you have a private parking spot outside your home?

The question asked respondents where they park their car at their residence. The responses were recoded into a binary variable indicating whether the household has access to a private parking spot.

- On the street, free unregulated (Not private)
- On the street, free regulated (SER, ORA, blue zone, green zone) (Not private)
- Own parking (Private)
- Resident parking (Private)
- Rented (Private)
- Space reserved for disabled people (Not private)
- Other (Not private)

- Do you have an electric/hybrid mode of transport?

The next dummy variable was created from the answers to the question which referred to whether the household has access to an electric or hybrid vehicle.

- Primary mode of transport.

- Renfe Commuter Rail (Rail)
- Intercity bus (Bus)
- Urban bus in another municipality (Bus)
- Metro (Metro)
- Light rail/tram (Other)
- Urban bus Madrid EMT (Bus)
- Other Renfe services (Other)
- Charter bus (Bus)
- Long-distance bus (Other)
- Taxi (Other)
- Private car driver (Car)
- Company car driver (Car)

- Rental car driver (without driver) (Car)
- Private car passenger (Car)
- Company car passenger (Car)
- Rental car passenger (with driver) (Car)
- Private motorcycle/moped (Other)
- Public motorcycle/moped (Other)
- Company motorcycle/moped (Other)
- Private bicycle (Other)
- Public bicycle (Other)
- Company bicycle (Other)
- Others (Other)
- Walking / on foot (Walking)

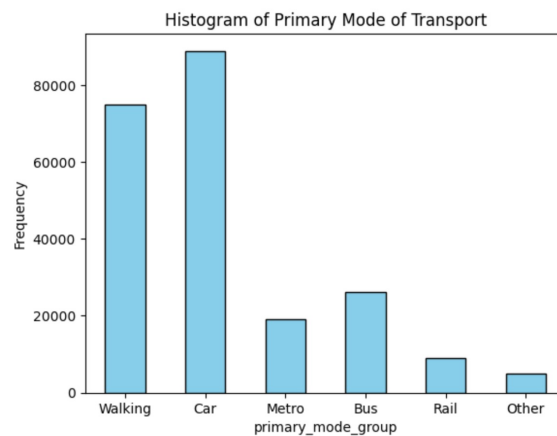


Figure 18: Primary mode of transport histogram

- Primary reason for the trip
 - Home (Other)
 - Work (Work)
 - Work-related management (Work)
 - Study (Study)
 - Shopping (Entertainment)
 - Medical (Other)
 - Accompanying another person (Accompanying another person)
 - Leisure (Entertainment)
 - Sports / Going for a walk (Entertainment)
 - Personal matters (Personal matters)
 - Another residence (Other)

- Others (Other)
- The dummies for rush hours are as follows:
 - Morning: 08:00 - 09:30
 - Midday: 13:30 - 15:00
 - Evening: 17:00 - 19:00

8.2 Latent Class Clustering

	LCC Clusters										
	2	3	4	5	6	7	8	9	10	11	12
AIC	6,184,102	5,879,108	5,723,874	5,591,763	5,533,679	5,489,735	5,433,677	5,408,861	5,340,682	5,329,475	5,289,674
BIC	6,185,762	5,881,604	5,727,205	5,595,929	5,538,681	5,495,573	5,440,350	5,416,370	5,349,025	5,338,655	5,299,689
CAIC	6,185,923	5,881,846	5,727,528	5,596,333	5,539,166	5,496,139	5,440,997	5,417,098	5,349,834	5,339,545	5,300,660

Table 8: Information criteria values (AIC, BIC, CAIC) for 2 to 12 clusters

	LCC Clusters										
	2	3	4	5	6	7	8	9	10	11	12
Entropy	1.000	0.979	0.977	0.986	0.974	0.974	0.981	0.988	0.983	0.970	0.980

Table 9: Entropy for 2 to 12 clusters

8.3 Evaluation of Clustering Methods by Classification

For a given class i , the precision and recall are defined as:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i}, \quad \text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \quad (26)$$

where TP_i is the True Positives for class i , FP_i the False Positives for class i , FN_i the False Negatives for class i . The F1 score for class i is then calculated as:

$$F1_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (27)$$

Weighted F1:

$$F1_{\text{weighted}} = \sum_{i=1}^C w_i \cdot F1_i \quad (28)$$

where, C is the number of classes, w_i the proportion of instances belonging to class i in the dataset: $w_i = \frac{n_i}{N}$, n_i the number of samples in class i , N the total number of samples.

8.4 K-Prototypes Algorithm

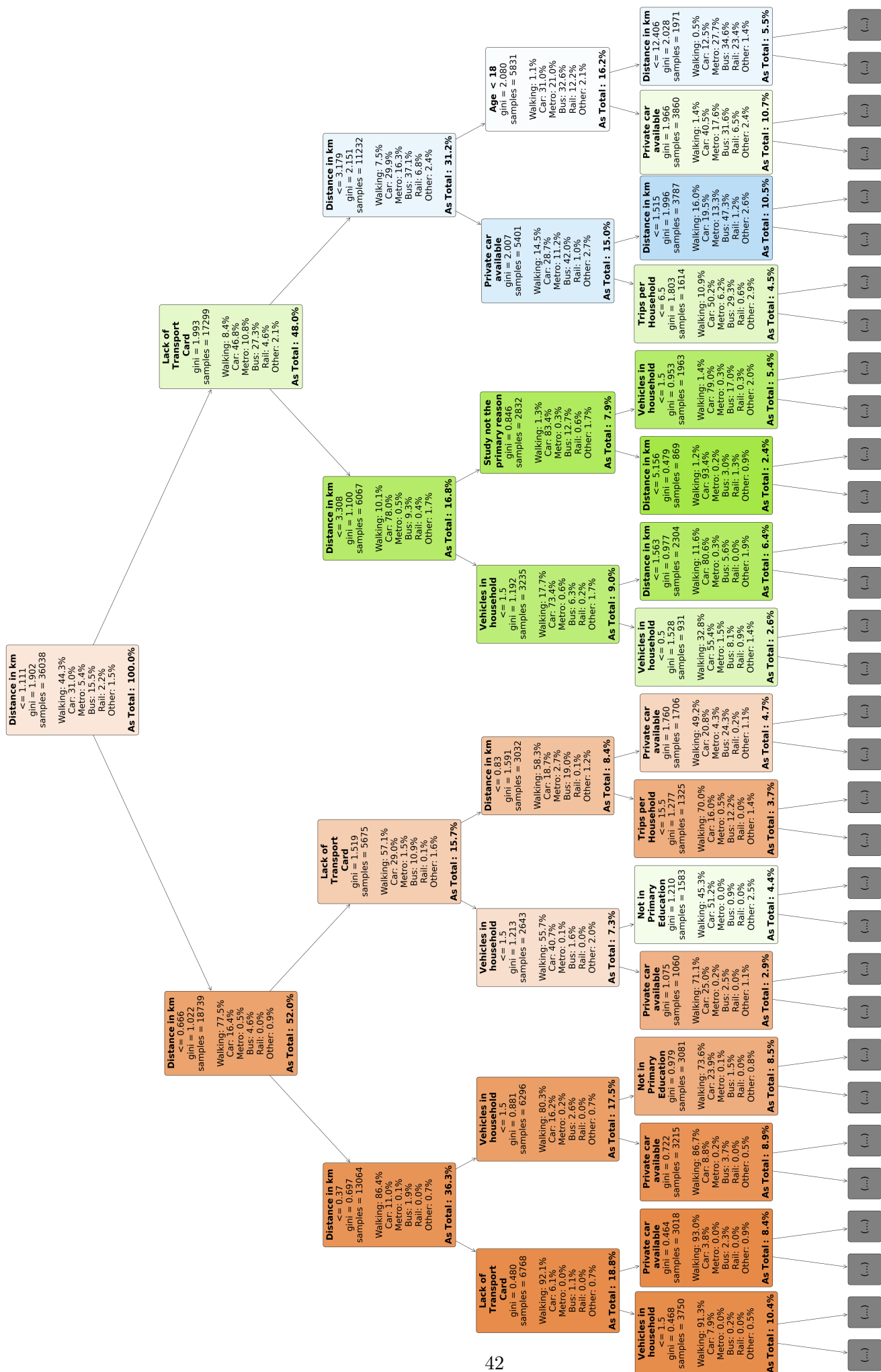
Step 1. Select k random data objects from the dataset as the initial prototypes of clusters.

- Step 2. For each of the data objects in the dataset, assign it to the nearest cluster based on the dissimilarity equation (5). At each assignment, update the corresponding prototype (update the mean for numerical attributes and the mode for categorical attributes).
- Step 3. For all data points, compute the dissimilarity with updated prototypes and reassign data points whose nearest prototype belongs to another cluster. After reassignments, update the prototypes.
- Step 4. If no data points were reassigned, the algorithm terminates. Otherwise, return to Step 3.

8.5 Classification and Regression Trees (CART)



Figure 19: Elderly Cluster Decision Tree Results: Transport Mode Choice



8.6 Weighted Average SHAP

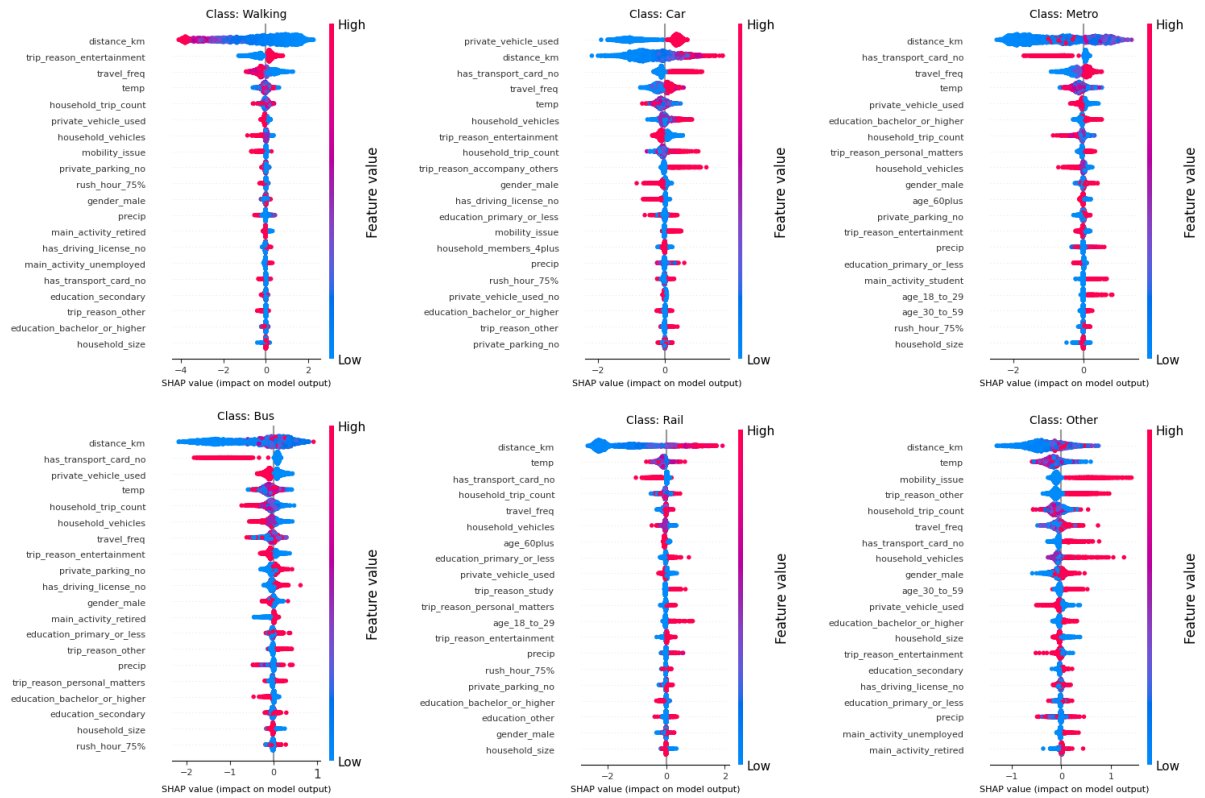


Figure 24: Weighted Average SHAP Plots For Each Mode of Transport For Cluster 1

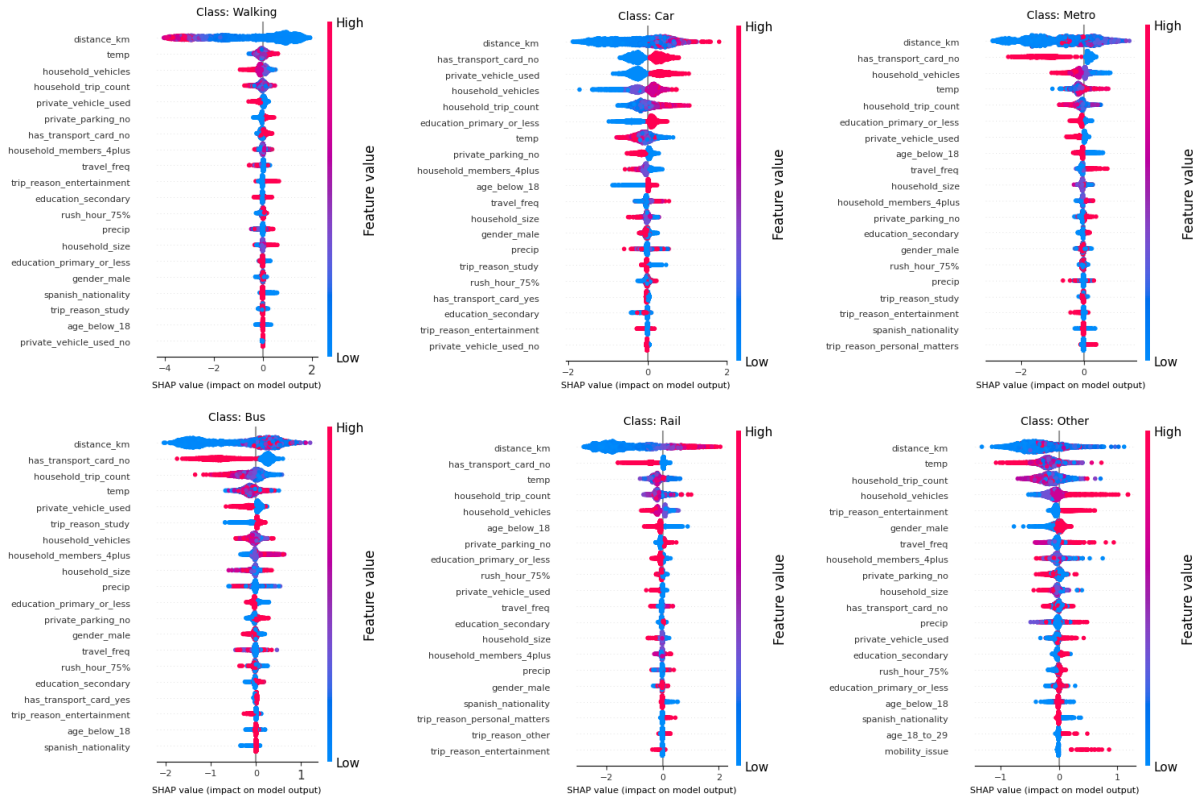


Figure 25: Weighted Average SHAP Plots For Each Mode of Transport For Cluster 2

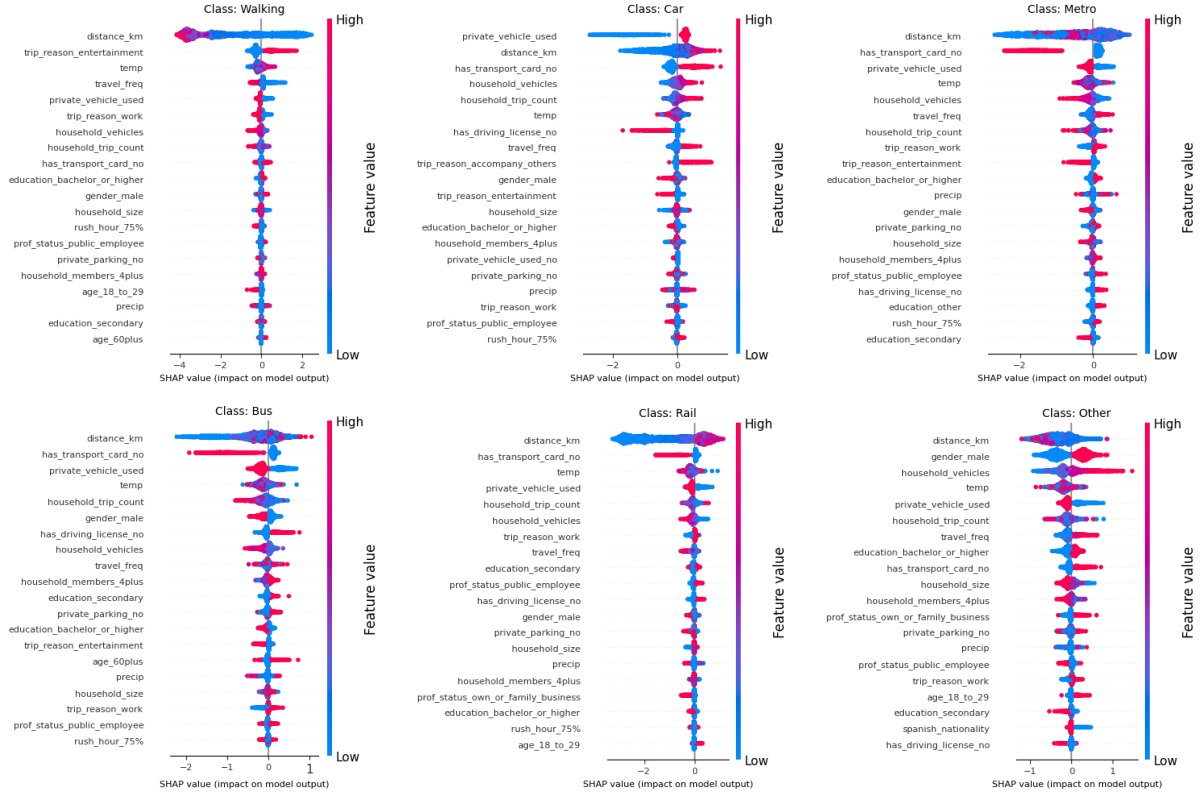


Figure 26: Weighted Average SHAP Plots For Each Mode of Transport For Cluster 3

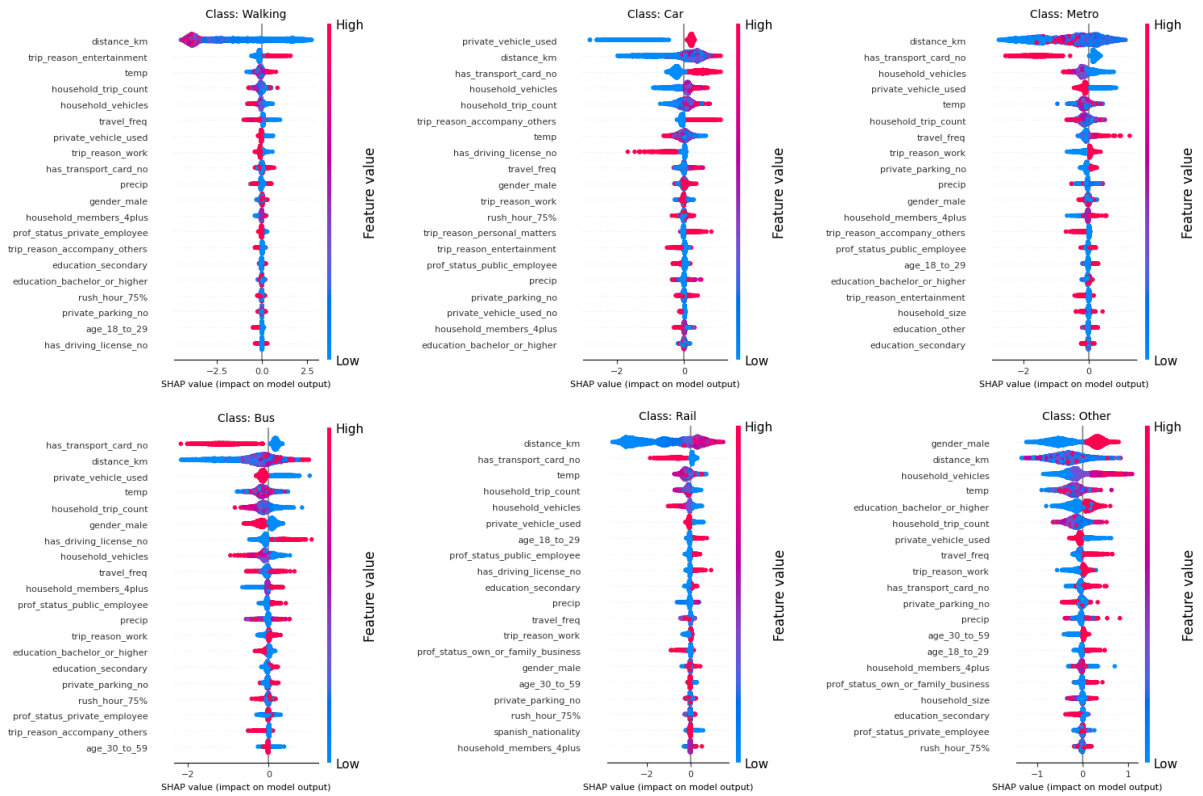


Figure 27: Weighted Average SHAP Plots For Each Mode of Transport For Cluster 4

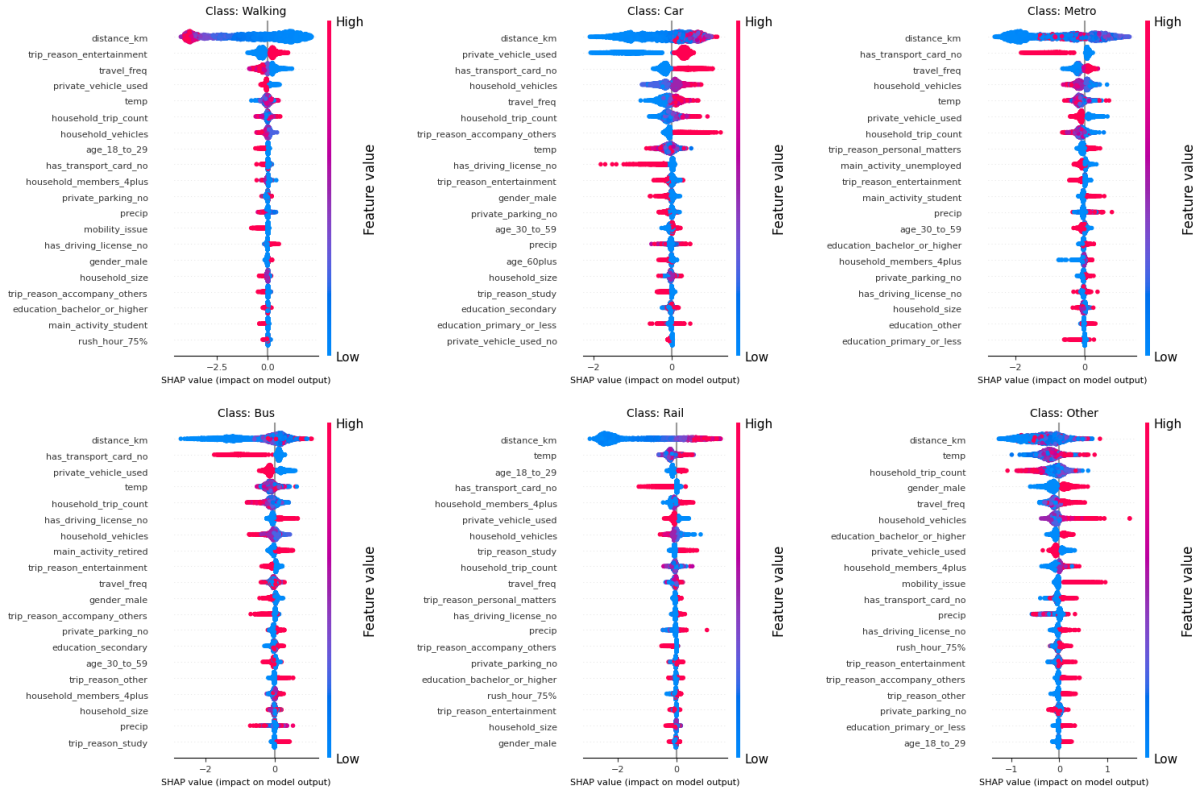


Figure 28: Weighted Average SHAP Plots For Each Mode of Transport For Cluster 5